

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent
Kind Code
Date of Patent
Inventor(s)

12417302
B1
September 16, 2025
Mehta; Neel et al.

Updating an edge device operating in a secure computing environment

Abstract

Described herein is a technique to update an edge device deployed in a secure computing network. A repository connected to a public network stores build contents configured to update software installed on the edge device; the public network is inaccessible to devices within the secure computing environment. A second device connected to the public network acquires the build contents in a signed lockbox file. An edge device management service generates a lockbox file containing the build contents and a trusted signer outside the secure computing network signs the lockbox file. The second device connects to secure computing network and establishes communications with the edge device. The edge device verifies the signed lockbox file provided by the second device. Upon verification, the edge device extracts the contents of the signed lockbox file and updates the software installed on the edge device.

Inventors: Mehta; Neel (Toronto, CA), Jacob; Joel (Etobicoke, CA), Ng; Anthony (Toronto, CA), Huang; William (Vaughan, CA), Quaresma; Rodrigo Paulo (Burlington, CA), Shum; Warren (Mississauga, CA), Yeung; Jonathan (Mississauga, CA), Sung; Alan (Richmond Hill, CA)

Applicant: SPLUNK INC. (San Francisco, CA)

Family ID: 1000006940611

Assignee: Splunk Inc. (San Francisco, CA)

Appl. No.: 18/162610

Filed: January 31, 2023

Publication Classification

Int. Cl.: G06F21/62 (20130101); G06F21/60 (20130101); G06F21/64 (20130101)

U.S. Cl.:

Field of Classification Search

CPC: G06F (21/6209); G06F (21/602); G06F (21/64)

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
8416954	12/2012	Raizen	726/2	G06F 16/13
2005/0144468	12/2004	Northcutt	713/192	H04L 63/0823
2015/0074409	12/2014	Reid	713/171	G06F 21/6218
2015/0161409	12/2014	Szebeni	713/165	H04L 63/08
2016/0277374	12/2015	Reid	N/A	H04L 63/0435
2018/0232526	12/2017	Reid	N/A	G06F 21/6218
2021/0014203	12/2020	Doshi	N/A	G06F 13/28
2021/0027559	12/2020	Fisher	N/A	G06Q 10/02
2021/0103530	12/2020	Don	N/A	G06F 12/1475
2021/0385069	12/2020	Reid	N/A	G06F 21/32
2022/0075695	12/2021	Khan	N/A	G06F 21/62
2022/0179974	12/2021	Hetzler	N/A	H04L 9/0894
2023/0127607	12/2022	Fernandez Orellana	726/11	H04L 63/0209
2023/0141909	12/2022	Truscott	726/23	H04L 63/02
2023/0315866	12/2022	Varteresian	726/26	G06F 21/602
2025/0028752	12/2024	Lauber	N/A	G06F 16/31

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
113052331	12/2020	CN	G06F 21/602
113596036	12/2020	CN	N/A

Primary Examiner: Keehn; Richard G

Attorney, Agent or Firm: Kilpatrick Townsend & Stockton LLP

Background/Summary

RELATED APPLICATIONS

(1) Any and all applications for which a foreign or domestic priority claim is identified in the Application Data Sheet as filed with the present application are incorporated by reference under 37 CFR 1.57 and made a part of this specification.

BACKGROUND

(2) Information technology (IT) environments can include diverse types of data systems that store large amounts of diverse data types generated by numerous devices. For example, a big data ecosystem may include databases such as MySQL and Oracle databases, cloud computing services such as Amazon web services (AWS), and other data systems that store passively or actively generated data, including machine-generated data (“machine data”). The machine data can include

log data, performance data, diagnostic data, metrics, tracing data, or any other data that can be analyzed to diagnose equipment performance problems, monitor user interactions, and to derive other insights.

(3) The large amount and diversity of data systems containing large amounts of structured, semi-structured, and unstructured data relevant to any search query can be massive and continues to grow rapidly. This technological evolution can give rise to various challenges in relation to collecting, managing, understanding, and effectively utilizing the data. To reduce the potentially vast amount of data that may be generated, some data systems pre-process data based on anticipated data analysis needs. In particular, specified data items may be extracted from the generated data and stored in a data system to facilitate efficient retrieval and analysis of those data items at a later time. At least some of the remainder of the generated data is typically discarded during pre-processing. Collecting and storing massive quantities of minimally processed or unprocessed data for later retrieval and analysis is becoming increasingly more feasible as new techniques are developed.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) Illustrative examples are described in detail below with reference to the following figures:
- (2) FIG. 1 illustrates a block diagram of an example data processing environment, in accordance with example implementations.
- (3) FIG. 2 illustrates a block diagram of an example data source, in accordance with example implementations.
- (4) FIG. 3 illustrates a block diagram of an example edge device, in accordance with example implementations.
- (5) FIG. 4 illustrates an example operation of an edge device upon receiving configuration data from an external sender, in accordance with example implementations.
- (6) FIG. 5 illustrates an example operation of an edge device upon receiving configuration data from an external sender, in accordance with example implementations.
- (7) FIG. 6 illustrates a block diagram of an example edge device within a data processing environment, in accordance with example implementations.
- (8) FIG. 7 illustrates a flowchart of an example process for delivering data from an edge device to an external recipient using a publish-subscribe protocol, in accordance with example implementations.
- (9) FIG. 8 illustrates an example operation of a repository manager generating a signed lockbox file for updating an edge device, in accordance with example implementations.
- (10) FIG. 9 illustrates an example operation of an edge device operating in a secure computing environment updating software installed in the edge device, in accordance with example implementations.
- (11) FIG. 10 illustrates a flowchart of an example process for updating software installed in an edge device operating in a secure computing environment, in accordance with example implementations.
- (12) FIG. 11 illustrates a perspective view of an example form factor of an edge device, in accordance with example implementations.
- (13) FIG. 12 illustrates a front view of the example form factor of the edge device, in accordance with example implementations.
- (14) FIG. 13 is a block diagram illustrating an example computing environment that includes a data intake and query system, in accordance with example implementations.
- (15) FIG. 14 is a block diagram illustrating in greater detail an example of an indexing system of a data intake and query system, in accordance with example implementations.

(16) FIG. 15 is a block diagram illustrating in greater detail an example of a search system of a data intake and query system, in accordance with example implementations.

(17) FIG. 16 illustrates an example of a self-managed network 1600 that includes a data intake and query system, in accordance with example implementations.

DETAILED DESCRIPTION

(18) Modern data centers and other computing environments can comprise anywhere from a few host computer systems to thousands of systems configured to process data, service requests from remote clients, and perform numerous other computational tasks. During operation, various components within these computing environments often generate significant volumes of machine data. Machine data is any data produced by a machine or component in an information technology (IT) environment that reflects activity in the IT environment. For example, machine data can be raw machine data that is generated by various components in IT environments, such as servers, sensors, routers, mobile devices, Internet of Things (IoT) devices, etc. Machine data can include system logs, network packet data, sensor data, application program data, error logs, stack traces, system performance data, etc. In general, machine data can also include performance data, diagnostic information, and many other types of data that can be analyzed to diagnose performance problems, monitor user interactions, and to derive other insights.

(19) A number of techniques are used to collect and analyze machine data. For example, edge devices coupled with sensors can be deployed within the IT environment to collect machine data and send the machine data to a data intake and query system. In such configurations, the edge devices and sensors function as data sources for the data intake and query system. The system may parse the machine data to produce events that each has a portion of machine data associated with a timestamp, and then store the events. The system enables users to run queries against the stored events to, for example, retrieve events that meet filter criteria specified in a query, such as criteria indicating certain keywords or having specific values in defined fields. Additional query terms can further process the event data, such as, by transforming the data, etc.

(20) A number of services execute at the edge device to manage the movement of the machine data that is captured by the sensors and is transmitted by the edge device to the data intake and query system. In some instances, the services may communicate with each other as well as with the sensors using a particular messaging protocol. Many of the conventional messaging protocols are insufficient for at least one of a number of reasons, including, (1) the messaging protocol may require excessive bandwidth due to, for example, the header size and/or the packet size being too large, (2) the messaging protocol may not be resource friendly by, for example, requiring too much memory and processing power, (3) the messaging protocol may not be sufficiently secure and may not allow for encrypting connections between entities, and (4) the messaging protocol may not provide sufficient reliability by, for example, allowing for quality of service (QOS) flags.

(21) These challenges and others can be addressed by the various examples of the present disclosure that, in some examples, provide for a messaging procedure employed at an edge device that enables efficient delivery of data to a data intake and query system. The edge device may include a system memory that has instructions stored therein for executing a message broker and a set of services. The message broker provides communication between a number of clients, which include the services running on the edge device as well as one or more sensors coupled to the edge device. The message broker may implement a topic-based publish-subscribe protocol in which messages are published by clients to certain topics and published messages are delivered to the clients that are subscribed to those topics. Each client may subscribe to one or more of the topics and the message broker may track these subscriptions by maintaining and updating a list of subscriptions.

(22) In some examples, a configuration file that contains configuration data may be loaded onto the edge device after it is received from an external sender. The configuration data, which may be unpackaged by a data streamer service running on the edge device, may indicate which topics the

data streamer service is to subscribe to and may further provide other instructions for modifying the operation of other services and sensors. In one example, the configuration data may include a request for anomaly data associated with a particular type of sensor data, and accordingly the data streamer service may subscribe to a topic for detected anomalies and an anomaly detection service may subscribe to a topic for that particular type of sensor data. Thereafter, the data streamer service may begin receiving published messages from the anomaly detection service that indicate whether an anomaly has been detected.

(23) In another example, the configuration data may include a request for a particular type of sensor data and a particular measurement rate for that type of sensor data (e.g., a request for temperature measurements at 0.5 Hz), and accordingly the data streamer service may subscribe to a topic for that particular type of sensor data and may further send a message to the sensor(s) to make measurements at the particular measurement rate. After subscribing to the topic for the particular type of sensor data, the data streamer service may begin receiving published messages that include sensor data and measurements for the particular type of sensor data. The sensor data may then be sent in the form of output data from the data streamer service to the data intake and query system or to some other external recipient over one or more networks.

(24) The figures herein follow a numbering convention in which the first digit or digits correspond to the figure number and the remaining digits identify an element or component in the figure. Similar elements or components between different figures may be identified by the use of similar digits. For example, 150 may reference element “50” in FIG. 1, and a similar element may be referenced as 250 in FIG. 2. As will be appreciated, elements shown in the various implementations herein can be added, exchanged, and eliminated so as to provide a number of additional implementations of the present disclosure.

(25) FIG. 1 illustrates a block diagram of an example data processing environment 100. In the illustrated example, the data processing environment 100 includes one or more data sources 102, a data intake and query system 110, and one or more computing devices 104 (alternatively referred to as “client devices” or “client computing devices”). Each of the data sources 102 may include an edge device 150 that is communicatively coupled with one or more sensors 152. In some examples, the data processing environment 100 may be alternatively referred to as a “computing environment”.

(26) The data intake and query system 110, edge devices 150, and computing devices 104 can communicate with each other via one or more networks, such as a local area network (LAN), wide area network (WAN), private or personal network, cellular networks, intranetworks, and/or internetworks using any of wired, wireless, terrestrial microwave, satellite links, etc., and may include the Internet. Although not explicitly shown in FIG. 1, it will be understood that a computing device 104 can communicate with an edge device 150 via one or more networks. For example, if the edge device 150 is configured as a web server and the computing device 104 is a laptop, the laptop can communicate with the web server to view a website.

(27) The computing devices 104 can correspond to distinct computing devices that can configure, manage, or sends queries to the system 110. Examples of the computing devices 104 may include, without limitation, smart phones, tablet computers, handheld computers, wearable devices, laptop computers, desktop computers, servers, portable media players, gaming devices, or other device that includes computer hardware (e.g., processors, non-transitory computer-readable media, etc.) and so forth. In certain cases, the computing devices 104 can include a hosted, virtualized, or containerized device, such as an isolated execution environment, that shares computing resources (e.g., processor, memory, etc.) of a particular machine with other isolated execution environments.

(28) The computing devices 104 can interact with the system 110 and/or the edge devices 150 in a variety of ways. For example, the computing devices 104 can communicate with the system 110 and/or the edge devices 150 over an Internet (Web) protocol, via a gateway, via a command line interface, via a software developer kit (SDK), a standalone application, etc. As another example,

the computing devices **104** can use one or more executable applications or programs to interface with the system **110**.

(29) The edge devices **150** can correspond to distinct computing devices or systems that include or have access to data that can be ingested, indexed, and/or searched by the system **110**. The edge devices **150** can include, but are not limited to, servers, routers, personal computers, mobile devices, internet of things (IoT) devices, or hosting devices, such as computing devices in a shared computing resource environment on which multiple isolated execution environment (e.g., virtual machines, containers, etc.) can be instantiated, or other computing devices in an IT environment (e.g., device that includes computer hardware, e.g., processors, non-transitory computer-readable media, etc.). In some examples, the edge devices **150** may receive the data from the sensors **152** that is to be processed by the system **110**. As such, each one of the edge devices **150** and its associated sensors **152** may constitute one of the data sources **102**.

(30) The types of data that are generated by each of the data sources **102** (and consequently by each of the edge devices **150**) can include machine data such as, for example and without limitation, server log files, activity log files, configuration files, messages, network packet data, performance measurements, sensor measurements, etc. In some cases, one or more applications executing on the edge devices **150** may generate various types of machine data during operation. For example, a web server application executing on one of the edge devices **150** may generate one or more web server logs detailing interactions between the web server and any number of the computing devices **104** or other devices.

(31) As another example, one of the edge devices **150** may be implemented as a router and may generate one or more router logs that record information related to network traffic managed by the router. As yet another example, a database server application executing on one of the edge devices **150** may generate one or more logs that record information related to requests sent from other devices (e.g., web servers, application servers, client devices, etc.) for data managed by the database server. Similarly, one of the edge devices **150** may generate and/or store computing resource utilization metrics, such as, but not limited to, CPU utilization, memory utilization, number of processes being executed, etc. Any one or any combination of the files or data generated in such cases can be used as a data source for the system **110**.

(32) As used herein, obtaining data from one of the data sources **102** may refer to communicating with one of the edge devices **150** to obtain data from the edge device **150** (e.g., from the sensors **152** associated with the edge device **150** or some other data streams or directories on the edge device **150**, etc.). For example, obtaining data from one of the data sources **102** may refer to requesting data from one of the edge devices **150** and/or receiving data from the edge device **150**. In some such cases, the edge device **150** can retrieve and return the requested data and/or the system **110** can retrieve the data from the edge device **150** (e.g., from a particular file stored on the edge device **150**).

(33) The data intake and query system **110** can ingest, index, and/or store data from heterogeneous data sources and/or edge devices **150**. For example, the system **110** can ingest, index, and/or store any type of machine data, regardless of the form of the machine data or whether the machine data matches or is similar to other machine data ingested, indexed, and/or stored by the system **110**. In some cases, the system **110** can generate events from the received data, group the events, and store the events in buckets. The system **110** can also search heterogeneous data that it has stored or search data stored by other systems (e.g., other system **110** systems or other non-system **110** systems). For example, in response to received queries, the system **110** can assign one or more components to search events stored in the storage system or search data stored elsewhere.

(34) As will be described herein in greater detail below, the system **110** can use one or more components to ingest, index, store, and/or search data. In some implementations, the system **110** is implemented as a distributed system that uses multiple components to perform its various functions. For example, the system **110** can include any one or any combination of an intake

system to ingest data, an indexing system to index the data, a storage system to store the data, and/or a query system (or search system) to search the data, etc. In some cases, the components of the system **110** are implemented as distinct computing devices having their own computer hardware (e.g., processors, non-transitory computer-readable media, etc.) and/or as distinct hosted devices (e.g., isolated execution environments) that share computing resources or hardware in a shared computing resource environment.

(35) The intake system can receive data from the edge devices **150**, perform one or more preliminary processing operations on the data, and communicate the data to the indexing system, query system, storage system, or to other systems (which may include, for example, data processing systems, telemetry systems, real-time analytics systems, data stores, databases, etc., any of which may be operated by an operator of the system **110** or a third party). Given the amount of data that can be ingested by the intake system, in some implementations, the intake system can include multiple distributed computing devices or components working concurrently to ingest the data. The preliminary processing operations performed by the intake system can include, but is not limited to, associating metadata with the data received from the edge devices **150**, extracting a timestamp from the data, identifying individual events within the data, extracting a subset of machine data for transmittal to the indexing system, enriching the data, etc.

(36) In some environments, a user of a system **110** may install and configure, on computing devices owned and operated by the user, one or more software applications that implement some or all of the components of the system **110**. For example, with reference to FIG. **1**, a user may install a software application on server computers owned by the user and configure each server to operate as one or more components of the intake system, indexing system, query system, shared storage system, or other components of the system **110**. This arrangement generally may be referred to as an “on-premises” solution. That is, the system **110** is installed and operates on computing devices directly controlled by the user of the system **110**. Some users may prefer an on-premises solution because it may provide a greater level of control over the configuration of certain aspects of the system (e.g., security, privacy, standards, controls, etc.). However, other users may instead prefer an arrangement in which the user is not directly responsible for providing and managing the computing devices upon which various components of system **110** operate.

(37) In certain examples, one or more of the components of the system **110** can be implemented in a shared computing resource environment. In this context, a shared computing resource environment or cloud-based service can refer to a service hosted by one more computing resources that are accessible to end users over a network, for example, by using a web browser or other application on a client device to interface with the remote computing resources. For example, a service provider may provide a system **110** by managing computing resources configured to implement various aspects of the system and by providing access to the system to end users via a network. Typically, a user may pay a subscription or other fee to use such a service. Each subscribing user of the cloud-based service may be provided with an account that enables the user to configure a customized cloud-based system based on the user's preferences.

(38) Implementing the system **110** in a shared computing resource environment can provide a number of benefits. In some cases, implementing the system **110** in a shared computing resource environment can make it easier to install, maintain, and update the components of the system **110**. For example, rather than accessing designated hardware at a particular location to install or provide a component of the system **110**, a component can be remotely instantiated or updated as desired. Similarly, implementing the system **110** in a shared computing resource environment or as a cloud-based service can make it easier to meet dynamic demand. For example, if the system **110** experiences significant load at indexing or search, additional compute resources can be deployed to process the additional data or queries. In an “on-premises” environment, this type of flexibility and scalability may not be possible or feasible.

(39) FIG. **2** illustrates a block diagram of an example data source **202**. In the illustrated example,

the data source **202** includes an edge device **250** that is communicatively coupled to a set of sensors **252**. The edge device **250** may include various hardware elements and software application programs that may be used by the hardware elements. For example, the edge device **250** may include a message broker **254** and a set of services **256** that are configured to run on the edge device **250**. For example, instructions for executing the message broker **254** and the services **256** may be stored on the system memory of the edge device **250** and, upon startup of the edge device **250**, these instructions may be sequentially loaded into one or more processors of the edge device **250** so that these programs are caused to run on the edge device **250** to carry out the functionalities described below.

(40) The message broker **254** is executed by the edge device **250** to provide communication between the various software and hardware entities within the data processing environment. For example, the message broker **254** may receive and send messages between several clients in accordance with a publish-subscribe network protocol. In some examples, the message broker **254** may implement a topic-based publish-subscribe protocol in which messages are published by clients on certain topics and the published messages are delivered by the message broker **254** to the clients that are subscribed to those topics. Clients may subscribe to one or more topics and the message broker **254** may track these subscriptions by maintaining a list of each subscription.

(41) The message broker **254** may directly or indirectly communicate with a number of clients, which may include one or more of the sensors **252** and one or more of the services **256**. Each of the clients may subscribe to one of a number of topics **258** that are maintained by the message broker **254**. The topics **258** may be a file or data structure that is prepopulated with the possible topics to which a client may subscribe or, in some examples, the topics **258** may be updated over time. For example, additional topics may be added to the topics **258** once the topic is first subscribed or published to, and topics may be removed from the topics **258** once the last client unsubscribes from the topic.

(42) The message broker **254** may maintain a list of subscriptions **262** to track the client subscriptions. In general, the list of subscriptions **262** may include one or more subscriptions that indicate which of the set of clients are subscribed to which of the topics **258**. The list of subscriptions **262** may be a file or data structure that is prepopulated with the subscriptions or, in some examples, is updated over time by, for example, adding a subscription each time a client subscribes to a topic to which the client was not previously subscribed, and removing a subscription each time a client unsubscribes from a topic. As described above, a client may subscribe to a topic that is previously listed in the topics **258** or is a new topic that may then be added to the topics **258**.

(43) In some examples, the message broker **254** may maintain a set of retained messages **264** that includes recent published messages received by the message broker **254**. In some examples, the retained messages **264** may be used to allow newly-subscribed clients to a topic to receive messages that were published prior to the clients being subscribed. In some examples, the publish-subscribe protocol may not require that at least one client must first be subscribed to a particular topic before any message can be published to that topic, and therefore a client that publishes a message has no guarantee that a subscribing client actually receives the message. By maintaining the retained messages **264**, clients may receive messages that they would otherwise have missed and, furthermore, a published message may be more likely to be received by a desired recipient. In various examples, the retained messages **264** may store the N most recent published messages, all messages published within the last T amount of time, or the N most recent published messages received within the last T amount of time, among other possibilities.

(44) As noted above, clients of the message broker **254** may include any of the sensors **252** and any of the services **256**. In various examples, one or more of the sensors **252** may be clients of the message broker **254** via direct communication with the message broker **254** or via one of the services **256** that may act as an intermediary between the message broker **254** and the sensors **252**.

For example, in the illustrated implementation, the sensors **252-1** and **252-2** may be clients of and may communicate directly with the message broker **254** while the sensors **252-3** and **252-4** may be clients of the message broker **254** and may communicate via service **256-4**, which may act as a sensor manager service that causes a connected sensor to perform various actions that change the operation of the connected sensor (e.g., turn on/off the sensor, increase/decrease the rate that sensor data is captured or transmitted).

(45) Further in the illustrated example, the services **256-1**, **256-2**, and **256-3** may be clients of the message broker **254** and may communicate directly with the message broker (e.g., by virtue of being executed on the same hardware). The services **256** may publish messages on certain topics and subscribe to certain topics so as to receive messages published to those topics. One or more of the services **256** may communicate with a data intake and query system **210** by, for example, receiving requests from the system **210** to subscribe to certain topics that the system **210** is interested in, and transmit messages published on those topics to the system **210**.

(46) The sensors **252** may include one or more of a variety of sensor types such as, without limitation, a light sensor, an image capture sensor, a sound sensor, a vibration sensor, an accelerometer, a gyroscope, a pressure sensor, a humidity sensor, a gas sensor, a location sensor, among other possibilities. While the sensors **252** are shown as being external to the edge device **250**, the sensors **252** may include a combination of internal and external sensors. For example, the sensors **252** may include an internal vibration sensor and/or an external vibration sensor that provide vibration measurements within the edge device **250** and of the external environment, respectively. External sensors may provide measurement data corresponding to a target device that is located within the data processing environment, such as a server computer, to which one or more of the sensors **252** are attached.

(47) FIG. 3 illustrates a block diagram of an example edge device **350**. In the illustrated example, the edge device **350** includes a message broker **354** and a set of services **356** that are configured to run on the edge device **350**. The message broker **354** may maintain a set of topics **358**, a list of subscriptions **362**, and a set of retained messages **364**. In the illustrated example, a set of topic IDs and client IDs are used by the message broker **354** to distinguish between different topics and clients, respectively.

(48) The illustrated example may represent the contents of the topics **358** and the list of subscriptions **362** at a particular point in time while the message broker **354** is running on the edge device **350**. The topics **358** include Topics T.1-T.12, which include topics for different types of sensor measurements, including Topic T.3 for temperature measurements, Topic T.4 for humidity measurements, and Topic T.5 for vibration measurements, as well as topics related to logs (Topic T.6) and anomalies (Topic T.7), among others. As described above, the number of topics in the topics **358** may increase or decrease when new topics are subscribed to or published on or when topics are no longer being subscribed to or published on.

(49) The list of subscriptions **362** includes subscriptions for clients corresponding to sensors as well as clients corresponding to the services **356**. In the illustrated example, the list of subscriptions **362** includes that Client Sensor. 1 is subscribed to Topics T.9 and T.10, that Client Service. 1 is subscribed to Topics T.3, T.4, and T.5, among others. As shown, multiple clients may be subscribed to a single topic, such as each of Clients Sensor.1, Sensor.2, and Sensor.3 being subscribed to Topics T.9 and T.10. Furthermore, sensor clients as well as service clients may be subscribed to a same topic, such as Clients Sensor. 1 and Service.5 being subscribed to Topic T.10.

(50) The illustrated example also shows several examples for services **356**, including an anomaly detection service **356-1**, a data streamer service **356-2**, a hardware control service **356-3**, a registration service **356-4**, a user interface (UI) service **356-5**, and a sensor management service **356-6**. In some examples, the anomaly detection service **356-1** may collect certain sensor data acquired by the sensors and detect anomalies associated with the sensor data. The anomaly detection service **356-1** may employ one or more machine learning (ML) models, where various

sensor data is inputted into one or more ML models to generate an output indicative of whether an anomaly was detected. For example, temperature data may be received by the anomaly detection service **356-1** and be inputted into a specific temperature ML model in order to identify anomalies and/or other alert conditions associated with a target operating temperature of a target device, the surrounding environment, or of the edge device **350** itself.

(51) In some examples, the data streamer service **356-2** may transmit data collected at the edge device **350** to a data intake and query system **310**. The data streamer service **356-2** may subscribe to one or more of the topics **358** in accordance with a configuration file or configuration data, which may be obtained (e.g., received) by the data streamer service **356-2** from an external device, such as the system **310**. For example, a configuration file received by the data streamer service **356-2** may indicate that certain sensor data (e.g., temperature data) is to be sent to the system **310**. The data streamer service **356-2** may then subscribe to the corresponding topic (e.g., Topic T.3) and relay data contained in any published messages back to the system **310**.

(52) The hardware control service **356-3** may control and manage the hardware components of the edge device **350**. The registration service **356-4** may register the edge device **350** with a remote application running on a remote device, allowing the remote device to send configuration data to the edge device **350** for modifying the functionality of one or more of the services **365**. The UI service **365-5** may manage the UI of the edge device **350** as well as any other I/O devices connected to or integrated with the edge device **350**. The sensor management service **356-6** may communicate with one or more connected sensors and perform various actions that change the operation of the sensors (e.g., increase the rate that certain sensor data is measured and/or transmitted).

(53) FIG. 4 illustrates an example operation of an edge device **450** upon receiving configuration data **472** from a sender that is external to the edge device **450**. In the illustrated example, the edge device **450** includes a message broker **454** and a set of services **456** including an anomaly detection service **456-1**, a data streamer service **456-2**, and a UI service **456-5** that are configured to run on the edge device **450**. The message broker **454** may maintain a set of topics **458**, a list of subscriptions **462**, and a set of retained messages **464**, and may facilitate the sending and receiving of messages **470** between clients. The message broker **454** may be communicatively coupled to the sensor **452**, which may be external or internal to the edge device **450**.

(54) In the illustrated example, the configuration data **472** is received by the edge device **450** (e.g., by the data streamer service **456-2**) from an external sender. In various examples, the data streamer service **456-2** may obtain the configuration data **472** using a variety of techniques. In one example, the data streamer service **456-2** may obtain the configuration data **472** directly from the external sender. In another example, a separate service running on the edge device **550** (referred to as the “pulse service”) may receive the configuration data **472** from the external sender and may publish a message containing the configuration data **472** on a particular topic for configuration data (such as Topic T.9) by sending the message to the message broker. The data streamer service **456-2**, which may have previously subscribed to the particular topic, may receive the published message from the message broker containing the configuration data. As such, in some examples, the data streamer service **456-2** may obtain the configuration data **472** via the message broker **354** by subscribing to a particular topic for configuration data.

(55) In various examples, the external sender may be a data intake and query system **410**, a computing or client device, a mobile device that is wirelessly connected to the edge device **450**, among other possibilities. In general, the configuration data **472** may include data for modifying the operation of clients of the message broker **454**, including the services **456** and the sensor **452**. The configuration data **472** may be received in the form of a configuration file.

(56) In the illustrated example, the configuration data **472** includes a request for temperature data (which is an example of a type of sensor data) and further specifies a particular temperature sampling rate (which is an example of a sensor sampling rate). The data streamer service **456-2**

may parse the configuration data 472 to identify the request for temperature data as well as the specified temperature sampling rate. In response to the data streamer service 456-2 obtaining the configuration data 472, the data streamer service 456-2 may send a message 470-1 to the message broker 454 to subscribe to Topic T.3 (i.e., the topic for temperature measurements). The data streamer service 456-2 may also cause the sensor 452 (e.g., through the sensor manager service) to modify its temperature sampling rate to the specified temperature sampling rate.

(57) In response to receiving the message 470-1, the message broker 454 may update the list of subscriptions 462 to indicate that the data streamer service 456-2 is subscribed to Topic T.3. In some examples, the message broker 454 may update the topics 458 to include Topic T.3 or, alternatively, prior to updating the list of subscriptions 462, the message broker 454 may verify that Topic T.3 is included in the topics 458. Optionally, further in response to the data streamer service 456-2 obtaining the configuration data 472, the anomaly detection service 456-1 and the UI service 456-5 may be caused to send messages 470-2 and 470-3, respectively, to the message broker 454 to subscribe to Topic T.3. In response to receiving the messages 470-2 and 470-3, the message broker 454 may update the list of subscriptions 462 to indicate that the anomaly detection service 456-1 and the UI service 456-5 are subscribed to Topic T.3.

(58) Thereafter, the sensor 452 may perform one or more temperature measurements at the specified temperature sampling rate. These measurements may be included in a message 470-4 (in the form of temperature data), which may be sent by the sensor 452 to the message broker 454 to publish the message 470-4 on Topic T.3. In response to receiving the message 470-4, the message broker 454 may examine the list of subscriptions 462 to identify which clients are subscribed to Topic T.3. After identifying that each of the services 456-1, 456-2, and 456-5 is subscribed to Topic T.3, the message broker 454 may send the messages 470-5, 470-6, and 470-7 to the services 456-1, 456-2, and 456-5, respectively, with each of these sent published messages including the same temperature data from the message 470-4.

(59) In response to receiving the message 470-6, the data streamer service 456-2 may prepare output data 474 that includes the temperature data and send the output data 474 to the system 410 for processing as described herein. In response to receiving the message 470-5, the anomaly detection service 456-1 may analyze the temperature data to possibly detect any anomalies in the data. In response to receiving the message 470-7, the UI service 456-5 may adjust the display of the edge device 450 to display the temperature measurements captured by the sensor 452.

(60) FIG. 5 illustrates an additional example operation of an edge device 550 upon receiving configuration data 572 from a sender that is external to the edge device 550. In the illustrated example, the edge device 550 includes a message broker 554 and a set of services 556 including an anomaly detection service 556-1, a data streamer service 556-2, and a hardware control service 556-5 that are configured to run on the edge device 550. The message broker 554 may maintain a set of topics 558, a list of subscriptions 562, and a set of retained messages 564, and may facilitate the sending and receiving of messages 570 between clients. The message broker 554 may be communicatively coupled to the sensor 552, which may be external or internal to the edge device 550.

(61) In the illustrated example, the configuration data 572 is received by the edge device 550 (e.g., obtained and/or received by the anomaly detection service 556-1 and the data streamer service 556-2) from an external sender. In various examples, the external sender may be a data intake and query system 510, a computing or client device, a mobile device that is wirelessly connected to the edge device 550, among other possibilities. In general, the configuration data 572 may include data for modifying the operation of clients of the message broker 554, including the services 556 and the sensor 552. The configuration data 572 may be received in the form of a configuration file.

(62) In the illustrated example, the configuration data 572 includes a request for anomaly data and further provides a ML model and associated weights to be used by the anomaly detection service 556-1. The configuration data 572 is obtained by the data streamer service 556-2 and optionally by

the anomaly detection service **556-1**. The data streamer service **556-2** may parse the configuration data **572** to identify the request for anomaly data as well as the ML model and associated weights. In response to the data streamer service **556-2** obtaining the configuration data **572**, the data streamer service **556-2** may send a message **570-1** to the message broker **554** to subscribe to Topic T.7 (i.e., the topic for anomaly data). The data streamer service **556-2** may also cause the anomaly detection service **556-1** to load the ML model and associated weights to be used for processing received sensor data.

(63) In response to receiving the message **570-1**, the message broker **554** may update the list of subscriptions **562** to indicate that the data streamer service **556-2** is subscribed to Topic T.7. In some examples, the message broker **554** may update the topics **558** to include Topic T.7 or, alternatively, prior to updating the list of subscriptions **562**, the message broker **554** may verify that Topic T.7 is included in the topics **558**. Further in response to the data streamer service **556-2** and/or the anomaly detection service **556-1** obtaining the configuration data **572**, the anomaly detection service **556-1** may send a message (not shown) to the message broker **554** to subscribe to one or more topics that are related to the input data for the ML model (e.g., topics related to temperature data or other sensor data). For example, the anomaly detection service **556-1** may subscribe to Topic T.3. Optionally, in response to the data streamer service **556-2** obtaining the configuration data **572**, the hardware control service **556-3** may be caused to send a message **570-2** to the message broker **554** to subscribe to Topic T.7. In response to receiving the messages **570-2**, the message broker **554** may update the list of subscriptions **562** to indicate that the hardware control service **556-3** is subscribed to Topic T.7.

(64) Thereafter, the sensor **552** may perform one or more temperature measurements. These measurements may be included in a message **570-3** (in the form of temperature data), which may be sent by the sensor **552** to the message broker **554** to publish the message **570-3** on Topic T.3. In response to receiving the message **570-3**, the message broker **554** may examine the list of subscriptions **562** to identify which clients are subscribed to Topic T.3. After identifying that the anomaly detection service **556-1** is subscribed to Topic T.3, the message broker **554** may send the messages **570-4** to the anomaly detection service **556-1**, with the sent published message including the same temperature data from the message **570-3**.

(65) In response to receiving the message **570-4**, anomaly detection service **556-1** may provide the temperature data as input to the ML model, which may produce an output that indicates whether or not an anomaly was detected. In the case where the output of the ML model indicates that an anomaly was detected based on the temperature data, the anomaly detection service **556-1** may generate a message **570-5** that includes anomaly data that identifies the detected anomaly. The message **570-5** is sent by the anomaly detection service **556-1** to the message broker **554** to publish the message **570-5** on Topic T.7.

(66) In response to receiving the message **570-5**, the message broker **554** may examine the list of subscriptions **562** to identify which clients are subscribed to Topic T.7. After identifying that each of the services **556-2** and **556-3** is subscribed to Topic T.7, the message broker **554** may send the messages **570-6** and **570-7** to the services **556-2** and **556-3**, respectively, with each of these sent published messages including the same anomaly data from the message **570-5**. In response to receiving the message **570-6**, the data streamer service **556-2** may prepare output data **574** that includes the anomaly data and send the output data **574** to the system **510** for processing as described herein. In response to receiving the message **570-7**, the hardware control service **556-3** may control one or more hardware elements of the edge device **550** based on the detected anomaly. For example, the hardware control service **556-3** may need to power off or reset the device, or cause warning indicators (e.g., LED lights) to be triggered to alert a user of the edge device **550** as to the detected anomaly.

(67) FIG. 6 illustrates a block diagram of an example edge device **650** within a data processing environment **600**. As shown, the data processing environment **600** may include, without limitation,

a data intake and query system **610** and an edge device **650** communicating with one another over one or more communications networks **678**. The edge device **650** may include, without limitation, a processor **682**, storage **684**, an input/output (I/O) device interface **688**, a network interface **690**, an interconnect **686**, and system memory **680**. The system memory **680** may include a message broker **654**, one or more services **656**, and one or more ML models **618**.

(68) In general, the processor **682** may retrieve and execute programming instructions stored in the system memory **680**, such as the message broker **654**, the services **656**, and ML models **618**, and any operating system stored therein. The processor **682** may be any technically-feasible form of a processing device configured to process data and execute program code. The processor **682** could be, for example, a central processing unit (CPU), a graphics processing unit (GPU), an application-specific integrated circuit (ASIC), a field-programmable gate array (FPGA), and so forth. The processor **682** stores and retrieves application data residing in the system memory **680**. The processor **682** is included to be representative of a single CPU, multiple CPUs, a single CPU having multiple processing cores, and the like. In operation, the processor **682** is the manager processor of the edge device **650**, controlling and coordinating operations of the other system components.

(69) The storage **684** may be a disk drive storage device. Although shown as a single unit, the storage **684** may be a combination of fixed and/or removable storage devices, such as fixed disc drives, floppy disc drives, tape drives, removable memory cards, or optical storage, network attached storage (NAS), or a storage area-network (SAN). The processor **682** may communicate to other computing devices and systems via the network interface **690**, where the network interface **690** is configured to transmit and receive data via the communications network **678**.

(70) The interconnect **686** facilitates transmission, such as of programming instructions and application data, between the processor **682**, the input/output (I/O) device interface **688**, the storage **684**, the network interface **690**, and the system memory **680**. The I/O device interface **688** is configured to transmit and receive data to and from one or more sensors **652** and I/O devices **622**. The I/O devices **622** may include one or more input devices (e.g., a keyboard, buttons, stylus, microphone, etc.) and/or one or more output devices (e.g., speaker, light-emitting diodes, etc.). In some instances, the I/O devices **622** includes a display device that displays an image and, in some examples, is integrated with the edge device **650**. In various examples, the display device **622** may be a liquid crystal display (LCD) display, organic light-emitting diode (OLED) display, or a digital light processing (DLP) display. In some instances, the sensors **652** may include a camera that acquires images via a lens and converts the images into digital form, which may then be displayed on the display device.

(71) The sensors **652** may include one or more of a variety of sensor types such as, without limitation, a light sensor, an image capture device (e.g., a camera), a sound sensor (e.g., microphone), a vibration sensor, one or more accelerometers (for measuring accelerations in one or more directions), one or more gyroscopes (for measuring rotations in one or more directions), a pressure sensor, a humidity sensor, a gas sensor (e.g., a CO.sub.2 sensor), a location sensor (e.g., a Global Navigation Satellite System (GNSS) receiver), among other possibilities. While the sensors **652** are shown as being external to the edge device **650**, the sensors **652** may be internal or external to edge device **650**. For example, the sensors **652** may include an internal vibration sensor and/or an external vibration sensor that provide vibration measurements within the edge device **650** and of the external environment, respectively. External sensors may provide measurement data corresponding to a target device, such as a server computer, to which one or more of the sensors **652** are attached.

(72) The services **656** may include a sensor manager service that may cause one or more of the sensors **652** to perform various actions that change the operation of the sensors **652** (e.g., increase rate that sensor data is transmitted) and/or the operation of the sensors **652** (e.g., turn on a camera and/or a microphone, etc.). The edge device **650** may execute the message broker **654** to

communicate with other devices within the data processing environment **600**. For example, the message broker **654** could receive sensor data from one or more of the sensors **652** and may send the data to the data intake and query system **610**. In various examples, the edge device **650** may process the received data. For example, the edge device **650** may retrieve one or more of the ML models **618** in order to process incoming data. In some implementations, the one or more ML models **618** are locally stored in the system memory **680** and locally updated at the edge device **650** without receiving updates from another device.

(73) FIG. 7 illustrates a flowchart of an example process **700** for delivering data from an edge device to an external recipient using a publish-subscribe protocol. The example process **700** can be implemented, for example, by a device that comprises a processor and a non-transitory computer-readable medium. The non-transitory computer readable medium can be storing instructions that, when executed by the processor, can cause the processor to perform the operations of the illustrated process **700**. Alternatively or additionally, the process **700** can be implemented using a non-transitory computer-readable medium storing instructions that, when executed by one or more processors, cause the one or more processors to perform the operations of the process **700** of FIG. 7.

(74) At step **702**, a list of subscriptions (e.g., list of subscriptions **262**, **362**, **462**, **562**) is maintained by a message broker (e.g., message brokers **254**, **354**, **454**, **554**, **654**) running on an edge device (e.g., edge devices **150**, **250**, **350**, **450**, **550**, **650**). The list of subscriptions may indicate which of a set of clients are subscribed to which of a set of topics (e.g., topics **258**, **358**, **458**, **558**). The set of clients may include one or more services (e.g., services **256**, **356**, **456**, **556**, **656**) running on the edge device.

(75) At step **704**, configuration data (e.g., configuration data **472**, **572**) is obtained by a data streamer service (e.g., data streamer services **356-2**, **456-2**, **556-2**) running on the edge device. The configuration data may include a request to receive a type of sensor data. The configuration data may alternatively or additionally include a request to receive anomaly data. The configuration data may alternatively or additionally include a ML model and associated weights to be used for generating the anomaly data. The configuration data may originate with a sender that is external to the edge device. The sender may be a user computing device (e.g., computing devices **104**), a data intake and query system (e.g., data intake and query systems **110**, **210**, **310**, **410**, **510**, **610**), or a server that is communicatively coupled to the edge device via a wireless connection.

(76) At step **706**, a first message (e.g., message **570-1**) subscribing to a first topic is sent from the data streamer service to the message broker. The first topic may correspond to the type of sensor data.

(77) At step **708**, the list of subscriptions is updated by the message broker to indicate that the data streamer service of the one or more services is subscribed to the first topic of the set of topics.

(78) At step **710**, a second message (e.g., message **570-3**) publishing on the first topic is received by the message broker. The second message may contain sensor data captured by a sensor (e.g., sensors **152**, **252**, **452**, **552**, **652**). The sensor data may include sensor measurements captured by the sensor. The set of clients may include the sensor. The sensor may generate the sensor data based on the sensor measurements, generate the second message to include the sensor data, and send the second message to the message broker. The second message may be received by the message broker via a sensor manager service (e.g., sensor manager services **356-6**) that is included in the one or more services. The sensor manager service may receive the sensor measurements from the sensor and generate the second message based on the sensor measurements.

(79) At step **712**, it is determined by the message broker that the data streamer service is subscribed to the first topic. The message broker may, for example, search the list of subscriptions to determine that the data streamer service is subscribed to the first topic. Step **712** may be performed in response to the message broker receiving the second message publishing on the first topic.

(80) At step **714**, the second message (e.g., message **470-6**) publishing on the first topic is sent from the message broker to the data streamer service. In response to receiving the second message,

the data streamer service may extract the sensor data from the second message.

(81) At step **716**, the sensor data is sent from the data streamer service to an external recipient. The sensor data may be sent in the form of output data (e.g., output data **474**, **574**) to the external recipient. The output data may further include anomaly data generated by the anomaly detection service. The external recipient may be the user computing device or the data intake and query system. In response to receiving the sensor data (and/or the anomaly data), the data intake and query system may parse, index, store, and search the sensor data (and/or the anomaly data) as described herein.

(82) Edge devices address several challenges related to transmitting physical sensor data into a data intake and query system, such as difficulties in integrating sensors that use proprietary communication formats or generate sensor data into repositories that cannot be accessed by certain devices due to security or compliance concerns. One such security limitation is a security policy that restricts third party services from accessing a secure computing network. Such a secure computing network substantially restricts or prevents devices within the secure computing network from communicating with devices outside of the secure computing network. A conventional solution for such situations involves installing customized edge devices that gathered sensor data within the secure computing network. Such edge devices would collect sensor data and transmit the sensor data to an on-premises data intake and query system located within the secure computing network.

(83) One drawback with conventional approaches for implementing edge devices within secure computing networks is that such approaches have difficulty managing complex structural changes or updates to software installed on the edge device. Conventional edge devices are configured to receive updates from a trusted source via over-the-air (OTA) communications to a device in a public network storing the applicable update files. However, an edge device deployed in a secure computing network cannot access the public network and is therefore unable to acquire the update files. As a result, a user must physically move the edge device outside the secure computing network in order to acquire the update files. Such cumbersome requirements result in users refraining from physically moving a given edge device for or moving a group of edge devices to an accessible location for each available update. As a result, edge devices deployed in secure computing networks run outdated or unsafe software, causing edge devices to be inefficient and vulnerable to attacks.

(84) These challenges and others can be addressed by the various examples of the present disclosure that, in some examples, provide for techniques to update an edge device deployed in a secure computing environment. Build contents, configured to update software installed on the edge device, are generated and stored in a repository connected to a public network. The public network is inaccessible to devices within the secure computing environment, including the edge device. A second device, such as a mobile device connected to the public network, acquires the build contents in the form of a signed lockbox file. An edge device management service generates a lockbox file containing the build contents and a trusted signer within the public network signs the lockbox file. The second device connects to secure computing network and establishes communications with the edge device. The edge device verifies the signed lockbox file provided by the second device. Upon verification, the edge device determines whether the update is a more recent software version than a software version currently installed on the edge device. Upon determining that the update is a more recent software version, the edge device extracts the contents of the signed lockbox file to update the software installed on the edge device.

(85) At least one technological advantage of the disclosed techniques relative to prior techniques is that an edge device can acquire, verify, and install software updates using build contents that would otherwise be inaccessible to the edge device deployed within the secure computing environment. Such techniques enable a user to provide updates to one or more edge devices deployed within a secure computing environment without requiring physical movement of each edge device to a

location where the edge device can access a public network to initiate an update process. Further, such techniques enable edge devices to update operations while complying with restrictive security and compliance limitations, expanding the number of environments that such edge devices can be implemented.

(86) FIG. 8 illustrates an example operation of a repository manager **802** generating a signed lockbox file **814** for updating an edge device, in accordance with example implementations. As shown, and without limitation, the network **800** includes a repository manager **802**, a pipeline **804**, a trusted signer **806**, edge device management services **808**, a data repository **812**, and a signed lockbox file **814**.

(87) In operation, the repository manager **802** prepares a set of update files associated with updating software installed on an edge device (e.g., edge devices **150**, **250**, **350**, **450**, **550**, **650**). For example, the repository manager **802** can compile a set of update files (e.g., build contents) into an image that enables the edge device to perform file system updates atomically. The repository manager **802** transmits the set of update files via the pipeline **804** to the edge device management services **808**, where the edge device management services **808** generate a lockbox file containing the set of update files. The edge device management services **808** transmit the lockbox file via the pipeline **804** to the trusted signer **806**. The trusted signer **806** signs the lockbox file to generate the signed lockbox file **814**. The trusted signer **806** transmits the signed lockbox file **814** to the data repository **812** via the pipeline **804**, where the data repository **812** stores the signed lockbox file **814** for subsequent acquisition by devices in the network **800**.

(88) The repository manager **802** is an artifact manager that houses and manages various files, such as binaries, packages, file groups, containers, and/or other components. In various implementations, the repository manager **802** is included (e.g., operates, executes, resides) in a first network and establishes communications with other components via the pipeline **804**. In various implementations, the repository manager **802** generates packages of build contents, including binaries and dependencies, and manages which packages are stored in the data repository **812**. For example, the repository manager **802** can manage which signed lockbox files **814** are stored in the data repository **812** for retrieval, replacing signed lockbox files associated with previous updates with the signed lockbox file **814** associated with the most recent update.

(89) In various implementations, the repository manager **802** manages the creation of the signed lockbox file **814** to enable secure offline updates of edge devices within a secure computing environment (e.g., a given edge device is not able to access the set of update files stored in the public network **800**). In some implementations, the signed lockbox file **814** is a collection of binary files, installation instructions, and/or software repository metadata. For example, the lockbox file can contain an image of file system software that modifies the operating system software installed on a given edge device, along with various dependencies (e.g., drivers, libraries, and settings) used during installation. In such instances, the repository manager **802** acts as a build server that compiles the image of the file system software.

(90) The pipeline **804** is a set of devices and/or software components that coordinate the generation and storage of the signed lockbox file **814**. In various implementations, the pipeline **804** is a continuous integration and continuous deployment (CI/CD) pipeline that includes components that execute a series of steps to deliver a new version of software. For example, the pipeline **804** coordinates the generation of the image of the file system software update by receiving build contents from the repository manager **802** and transmitting the build contents to the edge device management services **808**. In such instances, the edge device management services **808** generates a lockbox file containing the build contents and transmits the lockbox file to the pipeline **804**.

(91) The edge device management services **808** are services that manage one or more edge devices. In various implementations, the edge device management services **808** includes a sensor backend system that includes various services (e.g., registration services, sensor control services, service provisioning, reporting service, etc.) associated with the deployment and operation of edge devices.

In some implementations, the edge device management services **808** manage updates of edge devices by generating lockbox files containing content that enables an edge device to update software when offline. In some implementations, the edge device management services **808** verifies credentials provided by the pipeline **804** before generating the lockbox file containing the provided build contents.

(92) The trusted signer **806** is a user, device, or service that signs the lockbox file to generate the signed lockbox file **814**. In various implementations, the trusted signer **806** can be a cloud signing service that receives the lockbox file via the pipeline **804** and signs the lockbox file. In such instances, the trusted signer **806** can use various techniques, such as signing the lockbox file with a private key in a public/private key pair, to generate the signed lockbox file **814**. Upon the trusted signer **806** generating the signed lockbox file **814**, the repository manager **802** downloads the signed lockbox file **814** via the pipeline **804** and causes the signed lockbox file **814** to be stored in the data repository **812**.

(93) The data repository **812** is a device or service that stores the signed lockbox file **814**. In various implementations, the data repository **812** is a cloud storage service that provides object storage for various types of content items. In such instances, various types of devices can acquire the signed lockbox file **814** from the data repository **812**.

(94) FIG. 9 illustrates an example operation of an edge device operating in a secure computing environment updating software installed in the edge device, in accordance with example implementations. As shown, and without limitation, system **900** includes, without limitation, a secure computing network **902** and a public network **970**. The secure computing network **902** includes, without limitation, a computing device **904**, an edge device **950**, a trusted tunnel bridge **960**, and an on-premises data intake and query system **910**. The computing device **904** includes, without limitation, a signed lockbox file **914**. The edge device **950** includes, without limitation, a message broker **954** including topics **958** (e.g., an update topic **962** and a version updated topic **964**) and services **956** (e.g., a configuration service **956-1**, an update manager **956-2**, a service platform **956-3**, and a configuration subscription service **956-4**). The on-premises data intake and query system **910** includes, without limitation, a secure gateway **906** and an edge device management service **908**. The public network **970** includes, without limitation, the computing device **904** and the data repository **912**. The data repository **912** includes, without limitation, the signed lockbox file **914**.

(95) In operation, the computing device **904** facilitates the offline update of the edge device **950** using the signed lockbox file **914**. When the computing device **904** is within the public network **970**, the computing device **904** downloads a copy of the signed lockbox file **914-1** stored in the data repository **912**. Upon storing the copy as the signed lockbox file **914-2**, the computing device **904** moves into the secure computing network **902** to initiate the update process for the edge device **950**. When the computing device **904** is within the secure computing network **902**, the edge device **950** acquires the signed lockbox file **914-2** and verifies the signature of the signed lockbox file **914-2**. The edge device **950** uses the signed lockbox file **914-2** to update the applicable software. Upon updating the applicable software, the edge device **950** publishes a notification of the software update via a trusted tunnel bridge **960** to the on-premises data intake and query system **910**.

(96) In various implementations, the signed lockbox file **914-1** includes build contents to update software installed on the edge device **950**. Such build contents can include content items that enable new features, security upgrades, kernel patches, and/or improve overall experience and performance of the edge device **950**. In various implementations, when the edge device **950** is within the secure computing network **902**, various security and/or compliance requirements restrict access to the public network **970** or make the public network **970** wholly inaccessible. In such instances, the edge device **950** is unable to acquire the signed lockbox file **914-1** from the data repository **912**.

(97) The secure computing network **902** is a network substantially restricts or prevents devices

within the secure computing network **902** from communicating with devices outside of the secure computing network. For example, the computing device **904** and data repository **912** included in the public network **970** are inaccessible to the edge device **950** and the on-premises data intake and query system **910** within the secure computing network **902**. For example, the secure computing network **902** can be an air-gapped network that prevents wireless communications with devices outside of the secure computing network **902**. In some implementations, the secure computing network **902** can satisfy various security policies that specify rules for computer network access and/or comply with various security protocols imposed within the secure computing network. For example, the secure computing network can impose a security policy that includes one or more electronic security requirements (e.g., limiting inbound/outbound electronic access between devices outside the secure computing network **902** and devices within the secure computing network **902**, authentication of connectivity, etc.) and one or more physical security requirements (e.g., entities are to control physical access to devices within the secure computing network **902**).

(98) The computing device **904** is a device that can move between the public network **970** and the secure computing network **902**. For example, when the secure computing network **902** is an air gapped network, the computing device **904** can be a mobile device that physically moves from a first location where the public network **970** is accessible to a second location where the secure computing network **902** is accessible and the public network **970** is inaccessible. Alternatively or additionally, the computing device **904** can switch connections between networks **902**, **970**. For example, the computing device **904** can first connect to the public network **970**, then change connectivity to connect to the secure computing network **902**.

(99) When the computing device **904** is connected to the public network **970**, the computing device **904** acquires a copy of the signed lockbox file **914-1** (e.g., the signed lockbox file **914-2**) for use by the edge device **950** to perform a software update while the edge device **950** is within the secure computing network **902**. In some implementations, the computing device **904** provides credentials associated with the signed lockbox file **914-1** to the data repository **912** before the data repository **912** transmits a copy of the signed lockbox file **914-2**.

(100) When computing device **904** is connected to the secure computing network **902**, the computing device **904** provides the signed lockbox file **914-2** to the edge device **950**. In some implementations, a user manually starts the configuration service **956-1** on the edge device **950** to download the signed lockbox file **914-2**. In such instances, the configuration service **956-1** can display a graphical user interface at the edge device **950** to respond to manual inputs provided by the user to download, verify, and use the signed lockbox file **914-2** to update the edge device **950**. In some implementations, the configuration service **956-1** causes the edge device **950** to establish a direct connection between the computing device **904** and the edge device **950** (e.g., a connection via port **8000**). Alternatively or additionally, in some implementations, the computing device **904** uploads the signed lockbox file **914-2** to the configuration service **956-1** operating on the edge device **950**. In such instances, the edge device **950** can verify a set of credentials provided by the computing device **904** prior to downloading the signed lockbox file **914-2**. In various implementations, when the configuration service **956-1** downloads the signed lockbox file **914-2**, the configuration service **956-1** also publishes a message (e.g., message **570**) to the update topic **962**.

(101) In various implementations, the update manager **956-2** is a subscriber to the update topic **962**. When the configuration service **956-1** publishes a message to the update topic **962**, the update manager **956-2** processes the message in the update topic **962** and responds to the message by processing the signed lockbox file **914-1**. Processing the signed lockbox file **914-2** includes verifying the signed lockbox file **914-2**. For example, if the trusted signer **806** generated the signed lockbox file **914** using a private key in a private-public key pair, the update manager **956-2** can use the public key of the key pair to verify the signature of the signed lockbox file **914-2**. Upon verifying the signed lockbox file **914-2**, the update manager **956-2** locally stores the signed lockbox

file **914-2** and/or contents of the signed lockbox file **914-2**.

(102) In some implementations, the update manager **956-2** extracts the contents of the signed lockbox file **914-2**. In such instances, the extracted contents of the signed lockbox file **914-2** includes build contents (e.g., image of the software update, associated dependencies, etc.) and metadata associated with a specific version of the software. For example, the signed lockbox file **914-2** can include an internal manifest file that acts as a trusted source of version information. The edge device **950** uses the manifest file for version comparison against the version installed on the edge device **950** when determining whether to update the software using the build contents. For example, the update manager **956-2** can compare the version installed on the edge device **950** against the version specified in the manifest file to determine whether the build contents correspond to a newer software version. When the update manager **956-2** determines that the build contents correspond to a newer software version, the update manager **956-2** causes the build contents to be mounted within the service platform **956-3**.

(103) In various implementations, the service platform **956-3** updates applicable software operating on the edge device **950** using the build contents included in the signed lockbox file **914-2**. For example, the service platform **956-3** can include a pre-defined location within the system memory **680** that serves as a mount point for installation. For example, the update manager **956-2** can cause the build contents to be mounted at the pre-defined location and restart the edge device **950**. During the restart process, the service platform **956-3** processes the build contents at the mount point, causing the service platform **956-3** to update the software operating on the edge device **950**. For example, when the build contents are an image for file system software, the service platform **956-3** uses the mounted image to update the file system software operating on the edge device **950**.

(104) In various implementations, the update manager **956-3** monitors the update process being performed by the service platform **956-3**. In such instances, the update manager **956-3** publishes messages to cause the UI screen of the edge device **950** to provide notifications regarding the update process. When the service platform **956-3** completes the installation, the update manager **956-3** determines that installation is complete and publishes a message to the version updated topic **964**.

(105) In various implementations, the configuration subscription service **956-4** is a subscriber to the version updated topic **964** and processes the message published by the update manager **956-2**. In such instances, the configuration subscription service **956-4** transmits a message to the on-premises data intake and query system **910** confirming that the edge device **950** has successfully updated the software with the new software version. In some implementations, the configuration subscription service **956-4** establishes a trusted tunnel bridge **960** as a communication channel between the edge device **950** and the on-premises data intake and query system **910** to transmit the notification message.

(106) In various implementations, the edge device **950** uses the trusted tunnel bridge **960** to communicate with the on-premises data intake and query system **910** for other processes. For example, the edge device **950** can specify a specific operating (“hub”) state, such as a shutdown state, an update state, a sensing state etc. In such instances, the edge device **950** can periodically transmit a pulse request message containing the requested hub state to the on-premises data intake and query system **910**. In such instances, the on-premises data intake and query system **910** responds to the pulse request message by changing the hub state to the requested hub state.

(107) In some implementations, the on-premises data intake and query system **910** can register the edge device with the computing device **904**, account, and/or a specific user. For example, a user can maintain an account that includes a set of two or more edge devices **950**. In such instances, the on-premises data intake and query system **910** can remotely set one or more operational settings for each edge device **950** registered to the account. For example, the user can modify a service configuration detail (e.g., a data ingestion configuration, anomaly detection algorithm or threshold, etc.) for a first edge device **950-1**. Upon the first edge device **950-1** providing a notification to the

on-premises data intake and query system **910**, the edge device management service **908** in the on-premises data intake and query system **910** can transmit corresponding updates to one or more other edge devices (e.g., **950-2**, **950-3**, etc.) registered to the same account.

(108) In various implementations, the edge device **950** can communicate with one or more target devices using a version of a simple network management protocol (SNMP). In such instances, edge device **950** can monitor the one or more target devices based on polling data sent by the one or more target devices to the edge device **950**. In some implementations, a user can specify the Internet Protocol (IP) address of a given target device that is to be monitored. In some implementations, the user can also specify a specific SNMP protocol version and/or a management information base (MIB) file that supports the target device. In some implementations, the edge device **950** receives a selection of items to be monitored. Upon receiving the selection, the edge device **950** can periodically poll (e.g., every five minutes) the one or more target devices to retrieve information associated with the selected items. Upon receiving the information from the target devices, the edge device **950** can generate data that the on-premises data intake and query system **910** ingests as a series of event

(109) FIG. **10** illustrates a flowchart of an example process for updating software installed in an edge device operating in a secure computing environment, in accordance with example implementations. Although the method steps are described in conjunction with FIGS. **1-9**, persons of ordinary skill in the art will understand that any system configured to perform this method and/or other methods described herein, in any order, and in any combination not logically contradicted, is within the scope of the present invention.

(110) The example process **1000** can be implemented, for example, by a computing device that comprises a processor and a non-transitory computer-readable medium. The non-transitory computer readable medium can be storing instructions that, when executed by the processor, can cause the processor to perform the operations of the illustrated process **1000**. Alternatively or additionally, the process **1000** can be implemented using a non-transitory computer-readable medium storing instructions that, when executed by one or more processors, cause the one or more processors to perform the operations of the process **1000** of FIG. **1**.

(111) As shown by method **1000**, at step **1002**, where the edge device **950** receives a signed lockbox file **914** from a second device. In various implementations, a configuration service **956-1** operating on the edge device **950** receives a copy of a signed lockbox file **914-2** from a second device, such as the computing device **904** connected to the secure computing network **902**. In some implementations, the configuration service **956-1** causes the edge device **950** to establish a direct connection between the computing device **904** and the edge device **950** (e.g., a connection via port **8000**). Alternatively or additionally, in some implementations, the computing device **904** uploads the signed lockbox file **914-2** to the configuration service **956-1** operating on the edge device **950**. In such instances, the edge device **950** can verify a set of credentials provided by the computing device **904** prior to downloading the signed lockbox file **914-2**.

(112) In some implementations, the computing device **904** downloads the signed lockbox file **914** from the data repository **912** by accessing the public network **970**. In various implementations, the signed lockbox file **914-1** includes build contents to update software installed on the edge device **950**. Such build contents can include content items that enable new features, security upgrades, kernel patches, and/or improve overall experience and performance of the edge device **950**. In some implementations, the computing device **904** physically moves from a location where the public network **970** is accessible to a different location within the secure computing network **902** where the public network **970** is not accessible to the computing device **904**.

(113) At step **1004**, the edge device **950** determines whether to validate a signature of the signed lockbox file **914**. In various implementations, an update manager **956-2** operating on the edge device **950** processes the signed lockbox file **914** by verifying the signature of the signed lockbox file **914**. For example, the update manager **956-2** can use a public key of a key pair to determine

whether a trusted signer **806** validly signed the signed lockbox file **914** with the corresponding private key of the key pair. When the update manager **956-2** successfully validates the signature of the signed lockbox file **914**, the edge device **950** proceeds to step **1006**; otherwise, the update manager **956-2** determines that the signed lockbox file **914** cannot be validated and ends the method **1000**.

(114) At step **1006**, the edge device **950** extracts a set of update files from the signed lockbox file **914**. In various implementations, the update manager **956-2** extracts build contents and associated dependencies from the signed lockbox file **914**. For example, the signed lockbox file **914** can contain an image of file system software, an internal manifest file, and associated dependencies.

(115) At step **1008**, the edge device **950** determines whether the set of update files are for a newer version of the software than the version of the software installed on the edge device **950**. In various implementations, the update manager **956-2** compares versions of the software to determine whether to update the software. For example, the manifest file included in the signed lockbox file **914** can include version information corresponding to the build contents. In such instances, the update manager **956-2** compares the version associated with the build contents to the version of the software installed on the edge device **950**. When the update manager **956-2** determines that the version associated with the build contents is newer than the version installed on the edge device **950**, the edge device **950** proceeds to step **1010**. Otherwise, the update manager **956-2** determines that the version associated with the build contents is not newer than the version installed on the edge device **950** and ends the method **1000**.

(116) At step **1010**, the edge device **950** updates the edge device **950** using the set of update files included in the signed lockbox file **914**. In various implementations, a service platform **956-3** operating on the edge device **950** updates the software using the build contents included in the signed lockbox file **914**. For example, the service platform **956-3** can include a pre-defined location within the system memory **680** that serves as a mount point for installation. The update manager **956-2** causes the build contents to be mounted at the pre-defined location and restarts the edge device **950** for installation. During the restart process, the service platform **956-3** processes the build contents at the mount point, updating the software operating on the edge device **950**.

(117) In some implementations, the update manager **956-3** monitors the update process being performed by the service platform **956-3**. In such instances, the update manager **956-3** publishes notifications regarding the update process to a version updated topic **964** maintained by the edge device **950**. In such instances, a configuration subscription service **956-4** operating on the edge device **950** is a subscriber to the version updated topic **964** and processes the message published by the update manager **956-2**. The configuration subscription service **956-4** transmits a message to an on-premises data intake and query system **910** operating in the secure computing network **902** confirming that the edge device **950** has successfully updated the software with the new software version. In some implementations, the configuration subscription service **956-4** establishes a trusted tunnel bridge **960** as a communication channel between the edge device **950** and the on-premises data intake and query system **910** to transmit the notification message.

(118) FIG. **11** illustrates a perspective view of an example form factor of an edge device **1100**. The edge device **1100** includes a housing **1102** that encases one or more components (e.g., processor **682**, storage **684**, system memory **680**, one or more sensors **652**, etc.) of the edge device **1100**. In various implementations, the edge device **1100** may include optional ports, including one or more connection port(s) **1104**, USB port(s) **1106**, networking port(s) **1108**, and/or a power connector **1110** along one or more sides of housing **1102**.

(119) In some implementations, the connection ports **1104** can include one or more I/O ports that are configured to couple plugged-in devices (e.g., a sensor **652**, an input device **622-1**, an output device **622-2**, etc.) to the I/O devices interface **688**. The USB ports **1106** include one or more USB ports that are configured to couple plugged-in devices and/or power sources to the edge device **1100** (e.g., to the I/O devices interface **688**) via the USB protocol. The power connector **1110** is

configured to couple a power source (e.g., a wall power socket, an external battery) to the edge device **1100** via a wire or cable.

(120) In some implementations, the networking port(s) **1108** are configured to couple the network interface **690** to one or more network(s) **678** via a wire or cable, and any technically feasible protocol (e.g., Ethernet, Wi-Fi adapter, etc.). In various implementations, the edge device **1100** may use the networking port **1108** to attach to other networking ports, such as a switched port analyzer (SPAN) port of another network device, where the SPAN port takes a mirrored copy of network traffic and transmits the copy to a specific destination. In such instances, the edge device **1100** can measure specific network traffic metrics in order to monitor network characteristics associated with the secure computing network **902**. Additionally or alternatively, the edge device **1100** can implement the Message Queue Telemetry Transport (MQTT) protocol, Simple Network Management Protocol (SNMP), and/or other protocols to subscribe to specific message brokers and/or collect data about specific managed devices within a given data processing environment (e.g., secure computing network **902**).

(121) In some implementations, the housing **1102** is configured to be compliant with various environmental requirements. For example, the housing **1102** can be configured to be compliant with an ingress protection (IP-) rating of IP-**66**. A housing **1102** with an IP-**66** rating indicates that the edge device **1100** is dust tight (e.g., full protection against dust and other particulates, including a vacuum seal; tested against continuous airflow) and has high protection against moisture (e.g., protection against high-pressure jets of directed water from any angle; limited ingress with no harmful effects). In another example, the housing can be configured to be compliant with an IP-rating of IP-**67**. A housing **1102** with an IP-**67** rating also indicates that the edge device **1100** is dust tight and also indicates that the edge device **1100** has higher moisture protection against higher concentrations of moisture (e.g., protection against full immersion for up to 30 minutes at depths between 15 cm and 1 m; limited ingress permitted with no harmful effects). In some implementations, when the housing **1102** is configured to be IP-**66** or IP-**67** rated, one or more of ports **1104-1110** can be moved or removed in order to ensure that the housing **1102** remains dust tight and moisture resistant.

(122) FIG. **12** illustrates a front view of the example form factor of the edge device **1200**, in accordance with example implementations. The edge device **1200** includes a display device **1204** (e.g., an I/O device **622-3**) among its output devices **622**. The display device **1204** is included in the housing **1202** and is coupled to I/O devices interface **688** of the edge device **1200**.

(123) In some implementations, during operation, the edge device **1200** displays various information (e.g., via a user interface) on the display device **1204**. For example, the configuration service **956-1** operating on the edge device **1200** causes the display device **1204** to display prompts and notifications associated with updating the edge device **1200**.

(124) In some implementations, the edge device **1200** can include multiple sensors (e.g., sensors **652-1**, **652-2**) installed on a common printed circuit board (PCB). For example, the edge device **1200** can include a temperature sensor, a vibration sensor, a microphone, and an infrared sensor installed on a common printed circuit board. In such instances, the configuration service **956-1** and/or the service platform **956-3** can manage which sensors **652** are active. Additionally or alternatively, the edge device **1200** can include a second PCB that includes the processor. In such instances, the common PCB including the sensors can connect to the second PCB including the processor.

(125) Entities of various types, such as companies, educational institutions, medical facilities, governmental departments, and private individuals, among other examples, operate computing environments for various purposes. Computing environments, which can also be referred to as information technology environments, can include inter-networked, physical hardware devices, the software executing on the hardware devices, and the users of the hardware and software. As an example, an entity such as a school can operate a Local Area Network (LAN) that includes desktop

computers, laptop computers, smart phones, and tablets connected to a physical and wireless network, where users correspond to teachers and students. In this example, the physical devices may be in buildings or a campus that is controlled by the school. As another example, an entity such as a business can operate a Wide Area Network (WAN) that includes physical devices in multiple geographic locations where the offices of the business are located. In this example, the different offices can be inter-networked using a combination of public networks such as the Internet and private networks. As another example, an entity can operate a data center: a centralized location where computing resources are kept and maintained, and whose resources are accessible over a network. In this example, users associated with the entity that operates the data center can access the computing resources in the data center over public and/or private networks that may not be operated and controlled by the same entity. Alternatively or additionally, the operator of the data center may provide the computing resources to users associated with other entities, for example on a subscription basis. In both of these examples, users may expect resources to be available on demand and without direct active management by the user, a resource delivery model often referred to as cloud computing.

(126) Entities that operate computing environments need information about their computing environments. For example, an entity may need to know the operating status of the various computing resources in the entity's computing environment, so that the entity can administer the environment, including performing configuration and maintenance, performing repairs or replacements, provisioning additional resources, removing unused resources, or addressing issues that may arise during operation of the computing environment, among other examples. As another example, an entity can use information about a computing environment to identify and remediate security issues that may endanger the data, users, and/or equipment in the computing environment. As another example, an entity may be operating a computing environment for some purpose (e.g., to run an online store, to operate a bank, to manage a municipal railway, etc.) and information about the computing environment can aid the entity in understanding whether the computing environment is serving its purpose well.

(127) A data intake and query system can ingest and store data obtained from the components in a computing environment, and can enable an entity to search, analyze, and visualize the data. Through these and other capabilities, the data intake and query system can enable an entity to use the data for administration of the computing environment, to detect security issues, to understand how the computing environment is performing or being used, and/or to perform other analytics.

(128) FIG. 13 is a block diagram illustrating an example computing environment **1300** that includes a data intake and query system **1310**. The data intake and query system **1310** obtains data from a data source **1302** in the computing environment **1300**, and ingests the data using an indexing system **1320**. A search system **1360** of the data intake and query system **1310** enables users to navigate the indexed data. Though drawn with separate boxes, in some implementations the indexing system **1320** and the search system **1360** can have overlapping components. A computing device **1304**, running a network access application **1306**, can communicate with the data intake and query system **1310** through a user interface system **1314** of the data intake and query system **1310**. Using the computing device **1304**, a user can perform various operations with respect to the data intake and query system **1310**, such as administration of the data intake and query system **1310**, management and generation of “knowledge objects,” initiating of searches, and generation of reports, among other operations. The data intake and query system **1310** can further optionally include apps **1312** that extend the search, analytics, and/or visualization capabilities of the data intake and query system **1310**.

(129) The data intake and query system **1310** can be implemented using program code that can be executed using a computing device. A computing device is an electronic device that has a memory for storing program code instructions and a hardware processor for executing the instructions. The computing device can further include other physical components, such as a network interface or

components for input and output. The program code for the data intake and query system **1310** can be stored on a non-transitory computer-readable medium, such as a magnetic or optical storage disk or a flash or solid-state memory, from which the program code can be loaded into the memory of the computing device for execution. “Non-transitory” means that the computer-readable medium can retain the program code while not under power, as opposed to volatile or “transitory” memory or media that requires power in order to retain data.

(130) In various examples, the program code for the data intake and query system **1310** can execute on a single computing device, or may be distributed over multiple computing devices. For example, the program code can include instructions for executing both indexing and search components (which may be part of the indexing system **1320** and/or the search system **1360**, respectively), and can be executed on a computing device that also provides the data source **1302**. As another example, the program code can execute on one computing device, where the program code executes both indexing and search components, while another copy of the program code executes on a second computing device that provides the data source **1302**. As another example, the program code can execute only an indexing component or only a search component. In this example, a first instance of the program code that is executing the indexing component and a second instance of the program code that is executing the search component can be executing on the same computing device or on different computing devices.

(131) The data source **1302** of the computing environment **1300** is a component of a computing device that produces machine data. The component can be a hardware component (e.g., a microprocessor or a network adapter, among other examples) or a software component (e.g., a part of the operating system or an application, among other examples). The component can be a virtual component, such as a virtual machine, a virtual machine monitor (also referred as a hypervisor), a container, or a container orchestrator, among other examples. Examples of computing devices that can provide the data source **1302** include personal computers (e.g., laptops, desktop computers, etc.), handheld devices (e.g., smart phones, tablet computers, etc.), servers (e.g., network servers, compute servers, storage servers, domain name servers, web servers, etc.), network infrastructure devices (e.g., routers, switches, firewalls, etc.), and “Internet of Things” devices (e.g., vehicles, home appliances, factory equipment, etc.), among other examples. Machine data is electronically generated data that is output by the component of the computing device and reflects activity of the component. Such activity can include, for example, operation status, actions performed, performance metrics, communications with other components, or communications with users, among other examples. The component can produce machine data in an automated fashion (e.g., through the ordinary course of being powered on and/or executing) and/or as a result of user interaction with the computing device (e.g., through the user's use of input/output devices or applications). The machine data can be structured, semi-structured, and/or unstructured. The machine data may be referred to as raw machine data when the data is unaltered from the format in which the data was output by the component of the computing device. Examples of machine data include operating system logs, web server logs, live application logs, network feeds, metrics, change monitoring, message queues, and archive files, among other examples.

(132) As discussed in greater detail below, the indexing system **1320** obtains machine data from the data source **1302** and processes and stores the data. Processing and storing of data may be referred to as “ingestion” of the data. Processing of the data can include parsing the data to identify individual events, where an event is a discrete portion of machine data that can be associated with a timestamp. Processing of the data can further include generating an index of the events, where the index is a data storage structure in which the events are stored. The indexing system **1320** does not require prior knowledge of the structure of incoming data (e.g., the indexing system **1320** does not need to be provided with a schema describing the data). Additionally, the indexing system **1320** retains a copy of the data as it was received by the indexing system **1320** such that the original data is always available for searching (e.g., no data is discarded, though, in some examples, the indexing

system **1320** can be configured to do so).

(133) The search system **1360** searches the data stored by the indexing system **1320**. As discussed in greater detail below, the search system **1360** enables users associated with the computing environment **1300** (and possibly also other users) to navigate the data, generate reports, and visualize results in “dashboards” output using a graphical interface. Using the facilities of the search system **1360**, users can obtain insights about the data, such as retrieving events from an index, calculating metrics, searching for specific conditions within a rolling time window, identifying patterns in the data, and predicting future trends, among other examples. To achieve greater efficiency, the search system **1360** can apply map-reduce methods to parallelize searching of large volumes of data. Additionally, because the original data is available, the search system **1360** can apply a schema to the data at search time. This allows different structures to be applied to the same data, or for the structure to be modified if or when the content of the data changes. Application of a schema at search time may be referred to herein as a late-binding schema technique.

(134) The user interface system **1314** provides mechanisms through which users associated with the computing environment **1300** (and possibly others) can interact with the data intake and query system **1310**. These interactions can include configuration, administration, and management of the indexing system **1320**, initiation and/or scheduling of queries to the search system **1360**, receipt or reporting of search results, and/or visualization of search results. The user interface system **1314** can include, for example, facilities to provide a command line interface or a web-based interface.

(135) Users can access the user interface system **1314** using a computing device **1304** that communicates with data intake and query system **1310**, possibly over a network. A “user,” in the context of the implementations and examples described herein, is a digital entity that is described by a set of information in a computing environment. The set of information can include, for example, a user identifier, a username, a password, a user account, a set of authentication credentials, a token, other data, and/or a combination of the preceding. Using the digital entity that is represented by a user, a person can interact with the computing environment **1300**. For example, a person can log in as a particular user and, using the user's digital information, can access the data intake and query system **1310**. A user can be associated with one or more people, meaning that one or more people may be able to use the same user's digital information. For example, an administrative user account may be used by multiple people who have been given access to the administrative user account. Alternatively or additionally, a user can be associated with another digital entity, such as a bot (e.g., a software program that can perform autonomous tasks). A user can also be associated with one or more entities. For example, a company can have associated with it a number of users. In this example, the company may control the users' digital information, including assignment of user identifiers, management of security credentials, control of which persons are associated with which users, and so on.

(136) The computing device **1304** can provide a human-machine interface through which a person can have a digital presence in the computing environment **1300** in the form of a user. The computing device **1304** is an electronic device having one or more processors and a memory capable of storing instructions for execution by the one or more processors. The computing device **1304** can further include input/output (I/O) hardware and a network interface. Applications executed by the computing device **1304** can include a network access application **1306**, which can a network interface of the client computing device **1304** to communicate, over a network, with the user interface system **1314** of the data intake and query system **1310**. The user interface system **1314** can use the network access application **1306** to generate user interfaces that enable a user to interact with the data intake and query system **1310**. A web browser is one example of a network access application. A shell tool can also be used as a network access application. In some examples, the data intake and query system **1310** is an application executing on the computing device **1304**. In such examples, the network access application **1306** can access the user interface system **1314**

without needed to go over a network.

(137) The data intake and query system **1310** can optionally include apps **1312**. An app of the data intake and query system **1310** is a collection of configurations, knowledge objects (a user-defined entity that enriches the data in the data intake and query system **1310**), views, and dashboards that may provide additional functionality, different techniques for searching the data, and/or additional insights into the data. The data intake and query system **1310** can execute multiple applications simultaneously. Example applications include an information technology service intelligence application, which can monitor and analyze the performance and behavior of the computing environment **1300**, and an enterprise security application, which can include content and searches to assist security analysts in diagnosing and acting on anomalous or malicious behavior in the computing environment **1300**.

(138) Though FIG. **13** illustrates only one data source, in practical implementations, the computing environment **1300** contains many data sources spread across numerous computing devices. The computing devices may be controlled and operated by a single entity. For example, in an “on the premises” or “on-prem” implementation, the computing devices may physically and digitally be controlled by one entity, meaning that the computing devices are in physical locations that are owned and/or operated by the entity and are within a network domain that is controlled by the entity. In an entirely on-prem implementation of the computing environment **1300**, the data intake and query system **1310** executes on an on-prem computing device and obtains machine data from on-prem data sources. An on-prem implementation can also be referred to as an “enterprise” network, though the term “on-prem” refers primarily to physical locality of a network and who controls that location while the term “enterprise” may be used to refer to the network of a single entity. As such, an enterprise network could include cloud components.

(139) “Cloud” or “in the cloud” refers to a network model in which an entity operates network resources (e.g., processor capacity, network capacity, storage capacity, etc.), located for example in a data center, and makes those resources available to users and/or other entities over a network. A “private cloud” is a cloud implementation where the entity provides the network resources only to its own users. A “public cloud” is a cloud implementation where an entity operates network resources in order to provide them to users that are not associated with the entity and/or to other entities. In this implementation, the provider entity can, for example, allow a subscriber entity to pay for a subscription that enables users associated with subscriber entity to access a certain amount of the provider entity's cloud resources, possibly for a limited time. A subscriber entity of cloud resources can also be referred to as a tenant of the provider entity. Users associated with the subscriber entity access the cloud resources over a network, which may include the public Internet. In contrast to an on-prem implementation, a subscriber entity does not have physical control of the computing devices that are in the cloud, and has digital access to resources provided by the computing devices only to the extent that such access is enabled by the provider entity.

(140) In some implementations, the computing environment **1300** can include on-prem and cloud-based computing resources, or only cloud-based resources. For example, an entity may have on-prem computing devices and a private cloud. In this example, the entity operates the data intake and query system **1310** and can choose to execute the data intake and query system **1310** on an on-prem computing device or in the cloud. In another example, a provider entity operates the data intake and query system **1310** in a public cloud and provides the functionality of the data intake and query system **1310** as a service, for example under a Software-as-a-Service (SaaS) model. In this example, the provider entity can provision a separate tenant (or possibly multiple tenants) in the public cloud network for each subscriber entity, where each tenant executes a separate and distinct instance of the data intake and query system **1310**. In some implementations, the entity providing the data intake and query system **1310** is itself subscribing to the cloud services of a cloud service provider. As an example, a first entity provides computing resources under a public cloud service model, a second entity subscribes to the cloud services of the first provider entity and

uses the cloud computing resources to operate the data intake and query system **1310**, and a third entity can subscribe to the services of the second provider entity in order to use the functionality of the data intake and query system **1310**. In this example, the data sources are associated with the third entity, users accessing the data intake and query system **1310** are associated with the third entity, and the analytics and insights provided by the data intake and query system **1310** are for purposes of the third entity's operations.

(141) FIG. **14** is a block diagram illustrating in greater detail an example of an indexing system **1420** of a data intake and query system, such as the data intake and query system **1310** of FIG. **13**. The indexing system **1420** of FIG. **14** uses various methods to obtain machine data from a data source **1402** and stores the data in an index **1438** of an indexer **1432**. As discussed previously, a data source is a hardware, software, physical, and/or virtual component of a computing device that produces machine data in an automated fashion and/or as a result of user interaction. Examples of data sources include files and directories; network event logs; operating system logs, operational data, and performance monitoring data; metrics; first-in, first-out queues; scripted inputs; and modular inputs, among others. The indexing system **1420** enables the data intake and query system to obtain the machine data produced by the data source **1402** and to store the data for searching and retrieval.

(142) Users can administer the operations of the indexing system **1420** using a computing device **1404** that can access the indexing system **1420** through a user interface system **1414** of the data intake and query system. For example, the computing device **1404** can be executing a network access application **1406**, such as a web browser or a terminal, through which a user can access a monitoring console **1416** provided by the user interface system **1414**. The monitoring console **1416** can enable operations such as: identifying the data source **1402** for indexing; configuring the indexer **1432** to index the data from the data source **1402**; configuring a data ingestion method; configuring, deploying, and managing clusters of indexers; and viewing the topology and performance of a deployment of the data intake and query system, among other operations. The operations performed by the indexing system **1420** may be referred to as “index time” operations, which are distinct from “search time” operations that are discussed further below.

(143) The indexer **1432**, which may be referred to herein as a data indexing component, coordinates and performs most of the index time operations. The indexer **1432** can be implemented using program code that can be executed on a computing device. The program code for the indexer **1432** can be stored on a non-transitory computer-readable medium (e.g. a magnetic, optical, or solid state storage disk, a flash memory, or another type of non-transitory storage media), and from this medium can be loaded or copied to the memory of the computing device. One or more hardware processors of the computing device can read the program code from the memory and execute the program code in order to implement the operations of the indexer **1432**. In some implementations, the indexer **1432** executes on the computing device **1404** through which a user can access the indexing system **1420**. In some implementations, the indexer **1432** executes on a different computing device.

(144) The indexer **1432** may be executing on the computing device that also provides the data source **1402** or may be executing on a different computing device. In implementations wherein the indexer **1432** is on the same computing device as the data source **1402**, the data produced by the data source **1402** may be referred to as “local data.” In other implementations the data source **1402** is a component of a first computing device and the indexer **1432** executes on a second computing device that is different from the first computing device. In these implementations, the data produced by the data source **1402** may be referred to as “remote data.” In some implementations, the first computing device is “on-prem” and in some implementations the first computing device is “in the cloud.” In some implementations, the indexer **1432** executes on a computing device in the cloud and the operations of the indexer **1432** are provided as a service to entities that subscribe to the services provided by the data intake and query system.

- (145) For a given data produced by the data source **1402**, the indexing system **1420** can be configured to use one of several methods to ingest the data into the indexer **1432**. These methods include upload **1422**, monitor **1424**, using a forwarder **1426**, or using HyperText Transfer Protocol (HTTP **1428**) and an event collector **1430**. These and other methods for data ingestion may be referred to as “getting data in” (GDI) methods.
- (146) Using the upload **1422** method, a user can instruct the indexing system to **1402** to specify a file for uploading into the indexer **1432**. For example, the monitoring console **1416** can include commands or an interface through which the user can specify where the file is located (e.g., on which computing device and/or in which directory of a file system) and the name of the file. Once uploading is initiated, the indexer **1432** processes the file, as discussed further below. Uploading is a manual process and occurs when instigated by a user. For automated data ingestion, the other ingestion methods are used.
- (147) The monitor **1424** method enables the indexing system **1420** to monitor the data source **1402** and continuously or periodically obtain data produced by the data source **1402** for ingestion by the indexer **1432**. For example, using the monitoring console **1416**, a user can specify a file or directory for monitoring. In this example, the indexing system **1420** can execute a monitoring process that detects whenever data is added to the file or directory and causes the data to be sent to the indexer **1432**. As another example, a user can specify a network port for monitoring. In this example, a monitoring process can capture data received at or transmitting from the network port and cause the data to be sent to the indexer **1432**. In various examples, monitoring can also be configured for data sources such as operating system event logs, performance data generated by an operating system, operating system registries, operating system directory services, and other data sources.
- (148) Monitoring is available when the data source **1402** is local to the indexer **1432** (e.g., the data source **1402** is on the computing device where the indexer **1432** is executing). Other data ingestion methods, including forwarding and the event collector **1430**, can be used for either local or remote data sources.
- (149) A forwarder **1426**, which may be referred to herein as a data forwarding component, is a software process that sends data from the data source **1402** to the indexer **1432**. The forwarder **1426** can be implemented using program code that can be executed on the computer device that provides the data source **1402**. A user launches the program code for the forwarder **1426** on the computing device that provides the data source **1402**. The user can further configure the program code, for example to specify a receiver for the data being forwarded (e.g., one or more indexers, another forwarder, and/or another recipient system), to enable or disable data forwarding, and to specify a file, directory, network events, operating system data, or other data to forward, among other operations.
- (150) The forwarder **1426** can provide various capabilities. For example, the forwarder **1426** can send the data unprocessed or can perform minimal processing on the data. Minimal processing can include, for example, adding metadata tags to the data to identify a source, source type, and/or host, among other information, dividing the data into blocks, and/or applying a timestamp to the data. In some implementations, the forwarder **1426** can break the data into individual events (event generation is discussed further below) and send the events to a receiver. Other operations that the forwarder **1426** may be configured to perform include buffering data, compressing data, and using secure protocols for sending the data, for example.
- (151) Forwarders can be configured in various topologies. For example, multiple forwarders can send data to the same indexer. As another example, a forwarder can be configured to filter and/or route events to specific receivers (e.g., different indexers), and/or discard events. As another example, a forwarder can be configured to send data to another forwarder, or to a receiver that is not an indexer or a forwarder (such as, for example, a log aggregator).
- (152) The event collector **1430** provides an alternate method for obtaining data from the data

source **1402**. The event collector **1430** enables data and application events to be sent to the indexer **1432** using HTTP **1428**. The event collector **1430** can be implemented using program code that can be executing on a computing device. The program code may be a component of the data intake and query system or can be a standalone component that can be executed independently of the data intake and query system and operates in cooperation with the data intake and query system.

(153) To use the event collector **1430**, a user can, for example using the monitoring console **1416** or a similar interface provided by the user interface system **1414**, enable the event collector **1430** and configure an authentication token. In this context, an authentication token is a piece of digital data generated by a computing device, such as a server, that contains information to identify a particular entity, such as a user or a computing device, to the server. The token will contain identification information for the entity (e.g., an alphanumeric string that is unique to each token) and a code that authenticates the entity with the server. The token can be used, for example, by the data source **1402** as an alternative method to using a username and password for authentication.

(154) To send data to the event collector **1430**, the data source **1402** is supplied with a token and can then send HTTP **1428** requests to the event collector **1430**. To send HTTP **1428** requests, the data source **1402** can be configured to use an HTTP client and/or to use logging libraries such as those supplied by Java, JavaScript, and .NET libraries. An HTTP client enables the data source **1402** to send data to the event collector **1430** by supplying the data, and a Uniform Resource Identifier (URI) for the event collector **1430** to the HTTP client. The HTTP client then handles establishing a connection with the event collector **1430**, transmitting a request containing the data, closing the connection, and receiving an acknowledgment if the event collector **1430** sends one. Logging libraries enable HTTP **1428** requests to the event collector **1430** to be generated directly by the data source. For example, an application can include or link a logging library, and through functionality provided by the logging library manage establishing a connection with the event collector **1430**, transmitting a request, and receiving an acknowledgement.

(155) An HTTP **1428** request to the event collector **1430** can contain a token, a channel identifier, event metadata, and/or event data. The token authenticates the request with the event collector **1430**. The channel identifier, if available in the indexing system **1420**, enables the event collector **1430** to segregate and keep separate data from different data sources. The event metadata can include one or more key-value pairs that describe the data source **1402** or the event data included in the request. For example, the event metadata can include key-value pairs specifying a timestamp, a hostname, a source, a source type, or an index where the event data should be indexed. The event data can be a structured data object, such as a JavaScript Object Notation (JSON) object, or raw text. The structured data object can include both event data and event metadata. Additionally, one request can include event data for one or more events.

(156) In some implementations, the event collector **1430** extracts events from HTTP **1428** requests and sends the events to the indexer **1432**. The event collector **1430** can further be configured to send events or event data to one or more indexers. Extracting the events can include associating any metadata in a request with the event or events included in the request. In these implementations, event generation by the indexer **1432** (discussed further below) is bypassed, and the indexer **1432** moves the events directly to indexing. In some implementations, the event collector **1430** extracts event data from a request and outputs the event data to the indexer **1432**, and the indexer generates events from the event data. In some implementations, the event collector **1430** sends an acknowledgement message to the data source **1402** to indicate that the event collector **1430** has received a particular request from the data source **1402**, and/or to indicate to the data source **1402** that events in the request have been added to an index.

(157) The indexer **1432** ingests incoming data and transforms the data into searchable knowledge in the form of events. In the data intake and query system, an event is a single piece of data that represents activity of the component represented in FIG. **14** by the data source **1402**. An event can be, for example, a single record in a log file that records a single action performed by the

component (e.g., a user login, a disk read, transmission of a network packet, etc.). An event includes one or more fields that together describe the action captured by the event, where a field is a key-value pair (also referred to as a name-value pair). In some cases, an event includes both the key and the value, and in some cases the event includes only the value and the key can be inferred or assumed.

(158) Transformation of data into events can include event generation and event indexing. Event generation includes identifying each discrete piece of data that represents one event and associating each event with a timestamp and possibly other information (which may be referred to herein as metadata). Event indexing includes storing of each event in the data structure of an index. As an example, the indexer **1432** can include a parsing module **1434** and an indexing module **1436** for generating and storing the events. The parsing module **1434** and indexing module **1436** can be modular and pipelined, such that one component can be operating on a first set of data while the second component is simultaneously operating on a second set of data. Additionally, the indexer **1432** may at any time have multiple instances of the parsing module **1434** and indexing module **1436**, with each set of instances configured to simultaneously operate on data from the same data source or from different data sources. The parsing module **1434** and indexing module **1436** are illustrated to facilitate discussion, with the understanding that implementations with other components are possible to achieve the same functionality.

(159) The parsing module **1434** determines information about event data, where the information can be used to identify events within the event data. For example, the parsing module **1434** can associate a source type with the event data. A source type identifies the data source **1402** and describes a possible data structure of event data produced by the data source **1402**. For example, the source type can indicate which fields to expect in events generated at the data source **1402** and the keys for the values in the fields, and possibly other information such as sizes of fields, an order of the fields, a field separator, and so on. The source type of the data source **1402** can be specified when the data source **1402** is configured as a source of event data. Alternatively, the parsing module **1434** can determine the source type from the event data, for example from an event field or using machine learning.

(160) Other information that the parsing module **1434** can determine includes timestamps. In some cases, an event includes a timestamp as a field, and the timestamp indicates a point in time when the action represented by the event occurred or was recorded by the data source **1402** as event data. In these cases, the parsing module **1434** may be able to determine from the source type associated with the event data that the timestamps can be extracted from the events themselves. In some cases, an event does not include a timestamp and the parsing module **1434** determines a timestamp for the event, for example from a name associated with the event data from the data source **1402** (e.g., a file name when the event data is in the form of a file) or a time associated with the event data (e.g., a file modification time). As another example, when the parsing module **1434** is not able to determine a timestamp from the event data, the parsing module **1434** may use the time at which it is indexing the event data. As another example, the parsing module **1434** can use a user-configured rule to determine the timestamps to associate with events.

(161) The parsing module **1434** can further determine event boundaries. In some cases, a single line (e.g., a sequence of characters ending with a line termination) in event data represents one event while in other cases, a single line represents multiple events. In yet other cases, one event may span multiple lines within the event data. The parsing module **1434** may be able to determine event boundaries from the source type associated with the event data, for example from a data structure indicated by the source type. In some implementations, a user can configure rules the parsing module **1434** can use to identify event boundaries.

(162) The parsing module **1434** can further extract data from events and possibly also perform transformations on the events. For example, the parsing module **1434** can extract a set of fields for each event, such as a host or hostname, source or source name, and/or source type. The parsing

module **1434** may extract certain fields by default or based on a user configuration. Alternatively or additionally, the parsing module **1434** may add fields to events, such as a source type or a user-configured field. As another example of a transformation, the parsing module **1434** can anonymize fields in events to mask sensitive information, such as social security numbers or account numbers. Anonymizing fields can include changing or replacing values of specific fields. The parsing component **1434** can further perform user-configured transformations.

(163) The parsing module **1434** outputs the results of processing incoming event data to the indexing module **1436**, which performs event segmentation and builds index data structures.

(164) Event segmentation identifies searchable segments, which may alternatively be referred to as searchable terms or keywords, which can be used by the search system of the data intake and query system to search the event data. A searchable segment may be a part of a field in an event or an entire field. The indexer **1432** can be configured to identify searchable segments that are parts of fields, searchable segments that are entire fields, or both. The parsing module **1434** organizes the searchable segments into a lexicon or dictionary for the event data, with the lexicon including each searchable segment and a reference to the location of each occurrence of the searchable segment within the event data. As discussed further below, the search system can use the lexicon, which is stored in an index file **1446**, to find event data that matches a search query. In some implementations, segmentation can alternatively be performed by the forwarder **1426**.

Segmentation can also be disabled, in which case the indexer **1432** will not build a lexicon for the event data. When segmentation is disabled, the search system searches the event data directly.

(165) Building index data structures generates the index **1438**. The index **1438** is a storage data structure on a storage device (e.g., a disk drive or other physical device for storing digital data). The storage device may be a component of the computing device on which the indexer **1432** is operating (referred to herein as local storage) or may be a component of a different computing device (referred to herein as remote storage) that the indexer **1438** has access to over a network. The indexer **1432** can include more than one index and can include indexes of different types. For example, the indexer **1432** can include event indexes, which impose minimal structure on stored data and can accommodate any type of data. As another example, the indexer **1432** can include metrics indexes, which use a highly structured format to handle the higher volume and lower latency demands associated with metrics data.

(166) The indexing module **1436** organizes files in the index **1438** in directories referred to as buckets. The files in a bucket **1444** can include raw data files, index files, and possibly also other metadata files. As used herein, “raw data” means data as when the data was produced by the data source **1402**, without alteration to the format or content. As noted previously, the parsing component **1434** may add fields to event data and/or perform transformations on fields in the event data, and thus a raw data file **1448** can include, in addition to or instead of raw data, what is referred to herein as enriched raw data. The raw data file **1448** may be compressed to reduce disk usage. An index file **1446**, which may also be referred to herein as a “time-series index” or tsidx file, contains metadata that the indexer **1432** can use to search a corresponding raw data file **1448**. As noted above, the metadata in the index file **1446** includes a lexicon of the event data, which associates each unique keyword in the event data in the raw data file **1448** with a reference to the location of event data within the raw data file **1448**. The keyword data in the index file **1446** may also be referred to as an inverted index. In various implementations, the data intake and query system can use index files for other purposes, such as to store data summarizations that can be used to accelerate searches.

(167) A bucket **1444** includes event data for a particular range of time. The indexing module **1436** arranges buckets in the index **1438** according to the age of the buckets, such that buckets for more recent ranges of time are stored in short-term storage **1440** and buckets for less recent ranges of time are stored in long-term storage **1442**. Short-term storage **1440** may be faster to access while long-term storage **1442** may be slower to access. Buckets may move from short-term storage **1440**

to long-term storage **1442** according to a configurable data retention policy, which can indicate at what point in time a bucket is old enough to be moved.

(168) A bucket's location in short-term storage **1440** or long-term storage **1442** can also be indicated by the bucket's status. As an example, a bucket's status can be “hot,” “warm,” “cold,” “frozen,” or “thawed.” In this example, hot bucket is one to which the indexer **1432** is writing data and the bucket becomes a warm bucket when the index **1432** stops writing data to it. In this example, both hot and warm buckets reside in short-term storage **1440**. Continuing this example, when a warm bucket is moved to long-term storage **1442**, the bucket becomes a cold bucket. A cold bucket can become a frozen bucket after a period of time, at which point the bucket may be deleted or archived. An archived bucket cannot be searched. When an archived bucket is retrieved for searching, the bucket becomes thawed and can then be searched.

(169) The indexing system **1420** can include more than one indexer, where a group of indexers is referred to as an index cluster. The indexers in an index cluster may also be referred to as peer nodes. In an index cluster, the indexers are configured to replicate each other's data by copying buckets from one indexer to another. The number of copies of a bucket can be configured (e.g., three copies of each bucket must exist within the cluster), and indexers to which buckets are copied may be selected to optimize distribution of data across the cluster.

(170) A user can view the performance of the indexing system **1420** through the monitoring console **1416** provided by the user interface system **1414**. Using the monitoring console **1416**, the user can configure and monitor an index cluster, and see information such as disk usage by an index, volume usage by an indexer, index and volume size over time, data age, statistics for bucket types, and bucket settings, among other information.

(171) FIG. **15** is a block diagram illustrating in greater detail an example of the search system **1560** of a data intake and query system, such as the data intake and query system **1310** of FIG. **13**. The search system **1560** of FIG. **15** issues a query **1566** to a search head **1562**, which sends the query **1566** to a search peer **1564**. Using a map process **1570**, the search peer **1564** searches the appropriate index **1538** for events identified by the query **1566** and sends events **1578** so identified back to the search head **1562**. Using a reduce process **1582**, the search head **1562** processes the events **1578** and produces results **1568** to respond to the query **1566**. The results **1568** can provide useful insights about the data stored in the index **1538**. These insights can aid in the administration of information technology systems, in security analysis of information technology systems, and/or in analysis of the development environment provided by information technology systems.

(172) The query **1566** that initiates a search is produced by a search and reporting app **1516** that is available through the user interface system **1514** of the data intake and query system. Using a network access application **1506** executing on a computing device **1504**, a user can input the query **1566** into a search field provided by the search and reporting app **1516**. Alternatively or additionally, the search and reporting app **1516** can include pre-configured queries or stored queries that can be activated by the user. In some cases, the search and reporting app **1516** initiates the query **1566** when the user enters the query **1566**. In these cases, the query **1566** maybe referred to as an “ad-hoc” query. In some cases, the search and reporting app **1516** initiates the query **1566** based on a schedule. For example, the search and reporting app **1516** can be configured to execute the query **1566** once per hour, once per day, at a specific time, on a specific date, or at some other time that can be specified by a date, time, and/or frequency. These types of queries maybe referred to as scheduled queries.

(173) The query **1566** is specified using a search processing language. The search processing language includes commands that the search peer **1564** will use to identify events to return in the search results **1568**. The search processing language can further include commands for filtering events, extracting more information from events, evaluating fields in events, aggregating events, calculating statistics over events, organizing the results, and/or generating charts, graphs, or other visualizations, among other examples. Some search commands may have functions and arguments

associated with them, which can, for example, specify how the commands operate on results and which fields to act upon. The search processing language may further include constructs that enable the query **1566** to include sequential commands, where a subsequent command may operate on the results of a prior command. As an example, sequential commands may be separated in the query **1566** by a vertical line (“|” or “pipe”) symbol.

(174) In addition to one or more search commands, the query **1566** includes a time indicator. The time indicator limits searching to events that have timestamps described by the indicator. For example, the time indicator can indicate a specific point in time (e.g., 10:00:00 am today), in which case only events that have the point in time for their timestamp will be searched. As another example, the time indicator can indicate a range of time (e.g., the last 24 hours), in which case only events whose timestamps fall within the range of time will be searched. The time indicator can alternatively indicate all of time, in which case all events will be searched.

(175) Processing of the search query **1566** occurs in two broad phases: a map phase **1550** and a reduce phase **1552**. The map phase **1550** takes place across one or more search peers. In the map phase **1550**, the search peers locate event data that matches the search terms in the search query **1566** and sorts the event data into field-value pairs. When the map phase **1550** is complete, the search peers send events that they have found to one or more search heads for the reduce phase **1552**. During the reduce phase **1552**, the search heads process the events through commands in the search query **1566** and aggregate the events to produce the final search results **1568**.

(176) A search head, such as the search head **1562** illustrated in FIG. **15**, is a component of the search system **1560** that manages searches. The search head **1562**, which may also be referred to herein as a search management component, can be implemented using program code that can be executed on a computing device. The program code for the search head **1562** can be stored on a non-transitory computer-readable medium and from this medium can be loaded or copied to the memory of a computing device. One or more hardware processors of the computing device can read the program code from the memory and execute the program code in order to implement the operations of the search head **1562**.

(177) Upon receiving the search query **1566**, the search head **1562** directs the query **1566** to one or more search peers, such as the search peer **1564** illustrated in FIG. **15**. “Search peer” is an alternate name for “indexer” and a search peer may be largely similar to the indexer described previously. The search peer **1564** may be referred to as a “peer node” when the search peer **1564** is part of an indexer cluster. The search peer **1564**, which may also be referred to as a search execution component, can be implemented using program code that can be executed on a computing device. In some implementations, one set of program code implements both the search head **1562** and the search peer **1564** such that the search head **1562** and the search peer **1564** form one component. In some implementations, the search head **1562** is an independent piece of code that performs searching and no indexing functionality. In these implementations, the search head **1562** may be referred to as a dedicated search head.

(178) The search head **1562** may consider multiple criteria when determining whether to send the query **1566** to the particular search peer **1564**. For example, the search system **1560** may be configured to include multiple search peers that each have duplicative copies of at least some of the event data. In this example, the sending the search query **1566** to more than one search peer allows the search system **1560** to distribute the search workload across different hardware resources. As another example, search system **1560** may include different search peers for different purposes (e.g., one has an index storing a first type of data or from a first data source while a second has an index storing a second type of data or from a second data source). In this example, the search query **1566** may specify which indexes to search, and the search head **1562** will send the query **1566** to the search peers that have those indexes.

(179) To identify events **1578** to send back to the search head **1562**, the search peer **1564** performs a map process **1570** to obtain event data **1574** from the index **1538** that is maintained by the search

peer **1564**. During a first phase of the map process **1570**, the search peer **1564** identifies buckets that have events that are described by the time indicator in the search query **1566**. As noted above, a bucket contains events whose timestamps fall within a particular range of time. For each bucket **1544** whose events can be described by the time indicator, during a second phase of the map process **1570**, the search peer **1564** performs a keyword search **1574** using search terms specified in the search query **1566**. The search terms can be one or more of keywords, phrases, fields, Boolean expressions, and/or comparison expressions that in combination describe events being searched for. When segmentation is enabled at index time, the search peer **1564** performs the keyword search **1572** on the bucket's index file **1546**. As noted previously, the index file **1546** includes a lexicon of the searchable terms in the events stored in the bucket's raw data **1548** file. The keyword search **1572** searches the lexicon for searchable terms that correspond to one or more of the search terms in the query **1566**. As also noted above, the lexicon includes, for each searchable term, a reference to each location in the raw data **1548** file where the searchable term can be found. Thus, when the keyword search identifies a searchable term in the index file **1546** that matches query **1566**, the search peer **1564** can use the location references to extract from the raw data **1548** file the event data **1574** for each event that include the searchable term.

(180) In cases where segmentation was disabled at index time, the search peer **1564** performs the keyword search **1572** directly on the raw data **1548** file. To search the raw data **1548**, the search peer **1564** may identify searchable segments in events in a similar manner as when the data was indexed. Thus, depending on how the search peer **1564** is configured, the search peer **1564** may look at event fields and/or parts of event fields to determine whether an event matches the query **1566**. Any matching events can be added to the event data **#A74** read from the raw data **1548** file. The search peer **1564** can further be configured to enable segmentation at search time, so that searching of the index **1538** causes the search peer **1564** to build a lexicon in the index file **1546**.

(181) The event data **1574** obtained from the raw data **1548** file includes the full text of each event found by the keyword search **1572**. During a third phase of the map process **1570**, the search peer **1564** performs event processing **1576** on the event data **1574**, with the steps performed being determined by the configuration of the search peer **1564** and/or commands in the search query **1566**. For example, the search peer **1564** can be configured to perform field discovery and field extraction. Field discovery is a process by which the search peer **1564** identifies and extracts key-value pairs from the events in the event data **1574**. The search peer **1564** can, for example, be configured to automatically extract the first **100** fields (or another number of fields) in the event data **1574** that can be identified as key-value pairs. As another example, the search peer **1564** can extract any fields explicitly mentioned in the search query **1566**. The search peer **1564** can, alternatively or additionally, be configured with particular field extractions to perform.

(182) Other examples of steps that can be performed during event processing **1576** include: field aliasing (assigning an alternate name to a field); addition of fields from lookups (adding fields from an external source to events based on existing field values in the events); associating event types with events; source type renaming (changing the name of the source type associated with particular events); and tagging (adding one or more strings of text, or a "tags" to particular events), among other examples.

(183) The search peer **1564** sends processed events **1578** to the search head **1562**, which performs a reduce process **1580**. The reduce process **1580** potentially receives events from multiple search peers and performs various results processing **1582** steps on the events. The results processing **1582** steps can include, for example, aggregating the events from different search peers into a single set of events, deduplicating and aggregating fields discovered by different search peers, counting the number of events found, and sorting the events by timestamp (e.g., newest first or oldest first), among other examples. Results processing **1582** can further include applying commands from the search query **1566** to the events. The query **1566** can include, for example, commands for evaluating and/or manipulating fields (e.g., to generate new fields from existing

fields or parse fields that have more than one value). As another example, the query **1566** can include commands for calculating statistics over the events, such as counts of the occurrences of fields, or sums, averages, ranges, and so on, of field values. As another example, the query **1566** can include commands for generating statistical values for purposes of generating charts or graphs of the events.

(184) Through results processing **1582**, the reduce process **1580** produces the events found by processing the search query **1566**, as well as some information about the events, which the search head **1562** outputs to the search and reporting app **1516** as search results **1568**. The search and reporting app **1516** can generate visual interfaces for viewing the search results **1568**. The search and reporting app **1516** can, for example, output visual interfaces for the network access application **1506** running on a computing device **1504** to generate.

(185) The visual interfaces can include various visualizations of the search results **1568**, such as tables, line or area charts, Chloropleth maps, or single values. The search and reporting app **1516** can organize the visualizations into a dashboard, where the dashboard includes a panel for each visualization. A dashboard can thus include, for example, a panel listing the raw event data for the events in the search results **1568**, a panel listing fields extracted at index time and/or found through field discovery along with statistics for those fields, and/or a timeline chart indicating how many events occurred at specific points in time (as indicated by the timestamps associated with each event). In various implementations, the search and reporting app **1516** can provide one or more default dashboards. Alternatively or additionally, the search and reporting app **1516** can include functionality that enables a user to configure custom dashboards.

(186) The search and reporting app **1516** can also enable further investigation into the events in the search results **1516**. The process of further investigation may be referred to as drilldown. For example, a visualization in a dashboard can include interactive elements, which, when selected, provide options for finding out more about the data being displayed by the interactive elements. To find out more, an interactive element can, for example, generate a new search that includes some of the data being displayed by the interactive element, and thus may be more focused than the initial search query **1566**. As another example, an interactive element can launch a different dashboard whose panels include more detailed information about the data that is displayed by the interactive element. Other examples of actions that can be performed by interactive elements in a dashboard include opening a link, playing an audio or video file, or launching another application, among other examples.

(187) FIG. **16** illustrates an example of a self-managed network **1600** that includes a data intake and query system. “Self-managed” in this instance means that the entity that is operating the self-managed network **1600** configures, administers, maintains, and/or operates the data intake and query system using its own compute resources and people. Further, the self-managed network **1600** of this example is part of the entity's on-premise network and comprises a set of compute, memory, and networking resources that are located, for example, within the confines of a entity's data center. These resources can include software and hardware resources. The entity can, for example, be a company or enterprise, a school, government entity, or other entity. Since the self-managed network **1600** is located within the customer's on-prem environment, such as in the entity's data center, the operation and management of the self-managed network **1600**, including of the resources in the self-managed network **1600**, is under the control of the entity. For example, administrative personnel of the entity have complete access to and control over the configuration, management, and security of the self-managed network **1600** and its resources.

(188) The self-managed network **1600** can execute one or more instances of the data intake and query system. An instance of the data intake and query system may be executed by one or more computing devices that are part of the self-managed network **1600**. A data intake and query system instance can comprise an indexing system and a search system, where the indexing system includes one or more indexers **1620** and the search system includes one or more search heads **1660**.

(189) As depicted in FIG. 16, the self-managed network **1600** can include one or more data sources **1602**. Data received from these data sources may be processed by an instance of the data intake and query system within self-managed network **1600**. The data sources **1602** and the data intake and query system instance can be communicatively coupled to each other via a private network **1610**.

(190) Users associated with the entity can interact with and avail themselves of the functions performed by a data intake and query system instance using computing devices. As depicted in FIG. 16, a computing device **1604** can execute a network access application **1606** (e.g., a web browser), that can communicate with the data intake and query system instance and with data sources **1602** via the private network **1610**. Using the computing device **1604**, a user can perform various operations with respect to the data intake and query system, such as management and administration of the data intake and query system, generation of knowledge objects, and other functions. Results generated from processing performed by the data intake and query system instance may be communicated to the computing device **1604** and output to the user via an output system (e.g., a screen) of the computing device **1604**.

(191) The self-managed network **1600** can also be connected to other networks that are outside the entity's on-premise environment/network, such as networks outside the entity's data center. Connectivity to these other external networks is controlled and regulated through one or more layers of security provided by the self-managed network **1600**. One or more of these security layers can be implemented using firewalls **1612**. The firewalls **1612** form a layer of security around the self-managed network **1600** and regulate the transmission of traffic from the self-managed network **1600** to the other networks and from these other networks to the self-managed network **1600**.

(192) Networks external to the self-managed network can include various types of networks including public networks **1690**, other private networks, and/or cloud networks provided by one or more cloud service providers. An example of a public network **1690** is the Internet. In the example depicted in FIG. 16, the self-managed network **1600** is connected to a service provider network **1692** provided by a cloud service provider via the public network **1690**.

(193) In some implementations, resources provided by a cloud service provider may be used to facilitate the configuration and management of resources within the self-managed network **1600**. For example, configuration and management of a data intake and query system instance in the self-managed network **1600** may be facilitated by a software management system **1694** operating in the service provider network **1692**. There are various ways in which the software management system **1694** can facilitate the configuration and management of a data intake and query system instance within the self-managed network **1600**. As one example, the software management system **1694** may facilitate the download of software including software updates for the data intake and query system. In this example, the software management system **1694** may store information indicative of the versions of the various data intake and query system instances present in the self-managed network **1600**. When a software patch or upgrade is available for an instance, the software management system **1694** may inform the self-managed network **1600** of the patch or upgrade. This can be done via messages communicated from the software management system **1694** to the self-managed network **1600**.

(194) The software management system **1694** may also provide simplified ways for the patches and/or upgrades to be downloaded and applied to the self-managed network **1600**. For example, a message communicated from the software management system **1694** to the self-managed network **1600** regarding a software upgrade may include a Uniform Resource Identifier (URI) that can be used by a system administrator of the self-managed network **1600** to download the upgrade to the self-managed network **1600**. In this manner, management resources provided by a cloud service provider using the service provider network **1692** and which are located outside the self-managed network **1600** can be used to facilitate the configuration and management of one or more resources within the entity's on-prem environment. In some implementations, the download of the upgrades and patches may be automated, whereby the software management system **1694** is authorized to,

upon determining that a patch is applicable to a data intake and query system instance inside the self-managed network **1600**, automatically communicate the upgrade or patch to self-managed network **1600** and cause it to be installed within self-managed network **1600**. 1. In various implementations, a method comprises receiving, by an edge device in a secure computing network, a signed lockbox file from a second device, where the second device retrieves the signed lockbox file via a public network that is inaccessible from within the secure computing network, and the edge device receives the signed lockbox file from the second device when the second device is operating within the secure computing network, validating, by the edge device, a signature of the signed lockbox file, extracting a set of update files for the edge device from the signed lockbox file, and updating the edge device using the set of update files. 2. The method of clause 1, where the signed lockbox file is generated by an edge device management service operating in the public network. 3. The method of clause 1 or 2, where the signed lockbox file is signed with a first key by a signing service operating outside the secure computing network, and the edge device validates the signature using a second key corresponding to the first key. 4. The method of any of clauses 1-3, where the signed lockbox file is stored in a data repository outside of the secure computing network. 5. The method of any of clauses 1-4, further comprising publishing a message to an update topic included in a message broker of the edge device. 6. The method of any of clauses 1-5, where updating the edge device comprises responding to the message in the update topic by mounting the set of update files for access by an operating system of the edge device, restarting the edge device, and updating the edge device with the mounted set of files. 7. The method of any of clauses 1-6, where updating the edge device comprises comparing a version specified by a manifest included in the signed lockbox file to a version specified by locally-stored system files, where the edge device is updated with the set of update files when the version specified by the manifest is more recent than the version specified by the locally-stored system files. 8. The method of any of clauses 1-7, further comprising publishing a message to a message broker of the edge device. 9. The method of any of clauses 1-8, where the message broker is a publisher in a publish-subscribe network, a data intake and query system is a subscriber in the publish subscribe network, and the data intake and query system receives the message from the message broker via a trusted tunnel bridge. 10. The method of any of clauses 1-9, where the data intake and query system is included in the computing network and receives the message via a private trusted tunnel bridge. 11. The method of any of clauses 1-10, further comprising transmitting, by the edge device, a pulse to the data intake and query system, where the data intake and query system, in response to receiving the pulse, causes the edge device to change an operating state of the edge device. 12. The method of any of clauses 1-11, further comprising receiving, by the edge device from the data intake and query system, an update file associated with an operation setting, wherein the edge device is registered with a user modifying the operation setting, and updating, based on the update file, at least one service operating on the edge device. 13. The method of any of clauses 1-12, where the edge device receives the update file in response to the user modifying the at least one service operating on a separate edge device. 14. The method of any of clauses 1-13, where the public network is distinct from secure computing network, and the secure computing network satisfies a first security policy that includes at least one of a first set of electronic security requirements for the secure computing network or a first set of physical security requirements for one or more devices included in the computing network. 15. The method of any of clauses 1-14, where the edge device is enclosed in a housing that provides full protection against particulates and provides full protection against full immersion in a liquid at a depth of up to 1 meter. 16. The method of any of clauses 1-15, where the edge device comprises a printed circuit board (PCB), and two or more sensors are directly disposed on the PCB. 17. In various implementations, one or more non-transitory computer-readable media store instructions that, when executed by one or more processors, cause the one or more processors to perform the steps of receiving, by an edge device in a secure computing network, a signed lockbox file from a second device, where the second

device retrieves the signed lockbox file via a public network that is inaccessible from within the secure computing network, and the edge device receives the signed lockbox file from the second device when the second device is operating within the secure computing network, validating, by the edge device, a signature of the signed lockbox file, extracting a set of update files for the edge device from the signed lockbox file, and updating the edge device using the set of update files. 18. The one or more non-transitory computer-readable media of clause 17, where the signed lockbox file is signed with a first key by a signing service operating outside the secure computing network, and the edge device validates the signature using a second key corresponding to the first key. 19. In various implementations, an edge device comprises a memory storing instructions, and a processor coupled to the memory that executes the instructions by performing the steps of receiving, in a computing network, a signed lockbox file from a second device, where the second device retrieves the signed lockbox file via a public network that is inaccessible from within the computing network, and the edge device receives the signed lockbox file from the second device when the second device is operating within the computing network, validating a signature of the signed lockbox file, extracting a set of update files for the edge device from the signed lockbox file, and updating the edge device using the set of update files. 20. The edge device of clause 18, where the signed lockbox file is signed with a first key by a signing service operating outside the secure computing network, and the edge device validates the signature using a second key corresponding to the first key.

(195) Various examples and possible implementations have been described above, which recite certain features and/or functions. Although these examples and implementations have been described in language specific to structural features and/or functions, it is understood that the subject matter defined in the appended claims is not necessarily limited to the specific features or functions described above. Rather, the specific features and functions described above are disclosed as examples of implementing the claims, and other equivalent features and acts are intended to be within the scope of the claims. Further, any or all of the features and functions described above can be combined with each other, except to the extent it may be otherwise stated above or to the extent that any such implementations may be incompatible by virtue of their function or structure, as will be apparent to persons of ordinary skill in the art. Unless contrary to physical possibility, it is envisioned that (i) the methods/steps described herein may be performed in any sequence and/or in any combination, and (ii) the components of respective implementations may be combined in any manner.

(196) Processing of the various components of systems illustrated herein can be distributed across multiple machines, networks, and other computing resources. Two or more components of a system can be combined into fewer components. Various components of the illustrated systems can be implemented in one or more virtual machines or an isolated execution environment, rather than in dedicated computer hardware systems and/or computing devices. Likewise, the data repositories shown can represent physical and/or logical data storage, including, e.g., storage area networks or other distributed storage systems. Moreover, in some implementations the connections between the components shown represent possible paths of data flow, rather than actual connections between hardware. While some examples of possible connections are shown, any of the subset of the components shown can communicate with any other subset of components in various implementations.

(197) Examples have been described with reference to flow chart illustrations and/or block diagrams of methods, apparatus (systems), and computer program products. Each block of the flow chart illustrations and/or block diagrams, and combinations of blocks in the flow chart illustrations and/or block diagrams, may be implemented by computer program instructions. Such instructions may be provided to a processor of a general purpose computer, special purpose computer, specially-equipped computer (e.g., comprising a high-performance database server, a graphics subsystem, etc.) or other programmable data processing apparatus to produce a machine, such that

the instructions, which execute via the processor(s) of the computer or other programmable data processing apparatus, create means for implementing the acts specified in the flow chart and/or block diagram block or blocks. These computer program instructions may also be stored in a non-transitory computer-readable memory that can direct a computer or other programmable data processing apparatus to operate in a particular manner, such that the instructions stored in the computer-readable memory produce an article of manufacture including instruction means which implement the acts specified in the flow chart and/or block diagram block or blocks. The computer program instructions may also be loaded to a computing device or other programmable data processing apparatus to cause operations to be performed on the computing device or other programmable apparatus to produce a computer implemented process such that the instructions which execute on the computing device or other programmable apparatus provide steps for implementing the acts specified in the flow chart and/or block diagram block or blocks.

(198) In some implementations, certain operations, acts, events, or functions of any of the algorithms described herein can be performed in a different sequence, can be added, merged, or left out altogether (e.g., not all are necessary for the practice of the algorithms). In certain implementations, operations, acts, functions, or events can be performed concurrently, e.g., through multi-threaded processing, interrupt processing, or multiple processors or processor cores or on other parallel architectures, rather than sequentially.

Claims

1. A method comprising: receiving, by an edge device in a secure computing network, a signed lockbox file from a second device, wherein: the second device retrieves the signed lockbox file via a public network that is inaccessible from within the secure computing network, and the edge device receives the signed lockbox file from the second device when the second device is operating within the secure computing network; validating, by the edge device, a signature of the signed lockbox file; extracting a set of update files for the edge device from the signed lockbox file; and updating the edge device using the set of update files.
2. The method of claim 1, wherein the signed lockbox file is generated by an edge device management service operating in the public network.
3. The method of claim 1, wherein: the signed lockbox file is signed with a first key by a signing service operating outside the secure computing network; and the edge device validates the signature using a second key corresponding to the first key.
4. The method of claim 1, wherein the signed lockbox file is stored in a data repository outside of the secure computing network.
5. The method of claim 1, further comprising publishing a message to an update topic included in a message broker of the edge device.
6. The method of claim 5, wherein updating the edge device comprises: responding to the message in the update topic by mounting the set of update files for access by an operating system of the edge device, restarting the edge device, and updating the edge device with the set of update files that are mounted.
7. The method of claim 1, wherein updating the edge device comprises: comparing a version specified by a manifest included in the signed lockbox file to a version specified by locally-stored system files, wherein the edge device is updated with the set of update files when the version specified by the manifest is more recent than the version specified by the locally-stored system files.
8. The method of claim 1, further comprising publishing a message to a message broker of the edge device.
9. The method of claim 8, wherein: the message broker is a publisher in a publish-subscribe network; a data intake and query system is a subscriber in the publish subscribe network; and the

data intake and query system receives the message from the message broker via a trusted tunnel bridge.

10. The method of claim 9, wherein the data intake and query system is included in the secure computing network and receives the message via a private trusted tunnel bridge.

11. The method of claim 9, further comprising: transmitting, by the edge device, a pulse to the data intake and query system, wherein the data intake and query system, in response to receiving the pulse, causes the edge device to change an operating state of the edge device.

12. The method of claim 9, further comprising: receiving, by the edge device from the data intake and query system, an update file associated with an operation setting, wherein the edge device is registered with a user modifying the operation setting; and updating, based on the update file, at least one service operating on the edge device.

13. The method of claim 12, wherein the edge device receives the update file in response to the user modifying the at least one service operating on a separate edge device.

14. The method of claim 1, wherein: the public network is distinct from the secure computing network; and the secure computing network satisfies a first security policy that includes at least one of a first set of electronic security requirements for the secure computing network or a first set of physical security requirements for one or more devices included in the secure computing network.

15. The method of claim 1, wherein the edge device is enclosed in a housing that provides full protection against particulates and provides full protection against full immersion in a liquid at a depth of up to 1 meter.

16. The method of claim 1, wherein the edge device comprises: a printed circuit board (PCB); and two or more sensors are directly disposed on the PCB.

17. One or more non-transitory computer-readable media storing instructions that, when executed by one or more processors, cause the one or more processors to perform the steps of: receiving, by an edge device in a secure computing network, a signed lockbox file from a second device, wherein: the second device retrieves the signed lockbox file via a public network that is inaccessible from within the secure computing network, and the edge device receives the signed lockbox file from the second device when the second device is operating within the secure computing network; validating, by the edge device, a signature of the signed lockbox file; extracting a set of update files for the edge device from the signed lockbox file; and updating the edge device using the set of update files.

18. The one or more non-transitory computer-readable media of claim 17, wherein: the signed lockbox file is signed with a first key by a signing service operating outside the secure computing network; and the edge device validates the signature using a second key corresponding to the first key.

19. An edge device comprising: a memory storing instructions; and a processor coupled to the memory that executes the instructions by performing the steps of: receiving, in a computing network, a signed lockbox file from a second device, wherein: the second device retrieves the signed lockbox file via a public network that is inaccessible from within the computing network, and the edge device receives the signed lockbox file from the second device when the second device is operating within the computing network; validating a signature of the signed lockbox file; extracting a set of update files for the edge device from the signed lockbox file; and updating the edge device using the set of update files.

20. The edge device of claim 19, wherein: the signed lockbox file is signed with a first key by a signing service operating outside the computing network; and the edge device validates the signature using a second key corresponding to the first key.
